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Anatomic resolution of neurotransmitter-specific projections to the VTA reveals diversity of GABAergic inputs

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The ventral tegmental area (VTA) is important for reward processing and motivation. The anatomic organization of neurotransmitter-specific inputs to the VTA remains poorly resolved. In the present study, we mapped the major neurotransmitter projections to the VTA through cell-type-specific retrograde and anterograde tracing. We found that glutamatergic inputs arose from a variety of sources and displayed some connectivity biases toward specific VTA cell types. The sources of GABAergic projections were more widespread, displayed a high degree of differential innervation of subregions in the VTA and were largely biased toward synaptic contact with local GABA neurons. Inactivation of GABA release from the two major sources, locally derived versus distally derived, revealed distinct roles for these projections in behavioral regulation. Optogenetic manipulation of individual distal GABAergic inputs also revealed differential behavioral effects. These results demonstrate that GABAergic projections to the VTA are a major contributor to the regulation and diversification of the structure.

The VTA is a heterogeneous structure comprising dopamine-, GABA- and glutamate-producing neurons[1]. Adding to this complexity, neurons within this region have been shown to be multidimensional with respect to the neurotransmitters they can co-release[1]. Anatomically, the VTA can be subdivided into the rostral VTA (VTAR), rostral linear (RLi), caudal linear (CLi), interfascicular (IF), parabrachial pigmented (PBP) and paranigral (PN) regions[1]. In addition, the posterior VTA contains a high density of GABA-producing neurons and has been further delineated as the rostromedial tegmental area (RMTg)[2], or caudal tail of the VTA[3]. The cell types and subdivisions of the VTA have been proposed to serve different functions[4–7] through distinct input–output relationships[8–11]. However, cell-type-specific viral tracing studies have found a surprising lack of overall specificity of inputs on to the different cell types of the VTA[8,10,12,13].

Most projections to the VTA have largely been considered to be glutamatergic[14]. In this view, the diversity of glutamatergic inputs, together with a limited source of local and long-loop inhibitory inputs to the VTA, determines the functional diversity of the outputs[14,15]. However, accumulating evidence suggests that such an organization may be oversimplified. Ultrastructural analysis of synapses in the ventral midbrain revealed that symmetrical synapses, largely representing inhibitory synapses, outnumber asymmetrical excitatory synapses nearly tenfold[16]. Consistent with this, miniature inhibitory postsynaptic currents (mIPSCs) representing synaptic GABA connectivity are nearly ten times more frequent than miniature excitatory postsynaptic currents on to dopamine neurons[17]. Previous analysis of inputs to the VTA revealed a large number of projection neurons from a variety of anatomic sources that did not colocalize with glutamatergic markers, but the neurotransmitter phenotype of these projection neurons is not clear[15]. Of further note, nondopamine neurons in the VTA have almost three times more inhibitory synaptic inputs than dopamine neurons[18]. This disparity in synaptic connectivity could reflect a large number of inhibitory connections from local interneurons within the VTA, or from the RMTg. However, analyses of several individual projections to the VTA from other brain regions have found these to be largely GABAergic inputs that synapse predominantly on to GABA neurons[11,19–23]. Whether these observations are representative of the basic organization of inputs to the VTA or selective exceptions to the rule remains to be established.

In combination with anterograde tracing, synaptic connectivity analysis and gene inactivation, we find that GABAergic inputs represent the most widespread source of projections to the VTA. These projections largely innervate distinct subregions of the VTA, and nearly all inputs tested synapsed predominantly on to local GABAergic interneurons and weakly on to non-GABA neurons. Glutamatergic inputs are more limited in their sources and are less diversified in their innervation patterns, but different inputs display varying synaptic properties. Through genetic inactivation of functional GABA signaling from neurons originating within the ventral midbrain or projecting to the VTA from distal regions, we find that these two sources of inhibitory input differentially innervate dopamine and nondopamine neurons of the VTA and have distinct roles in reinforcement learning and social behavior. Furthermore, optogenetic activation and inhibition of distal GABA inputs with varying VTA innervation patterns revealed differential effects on reinforcement and social behaviors. Collectively, these data provide new insights into the basic organizational principles of the VTA, and represent an anatomic and functional guide for further dissection of dopamine- and GABA-dependent processes.

Results

Anatomic organization of neurotransmitter inputs to the VTA. The major neurotransmitter inputs to the VTA are GABA, glutamate, serotonin and acetylcholine[14]. To establish the sources of these

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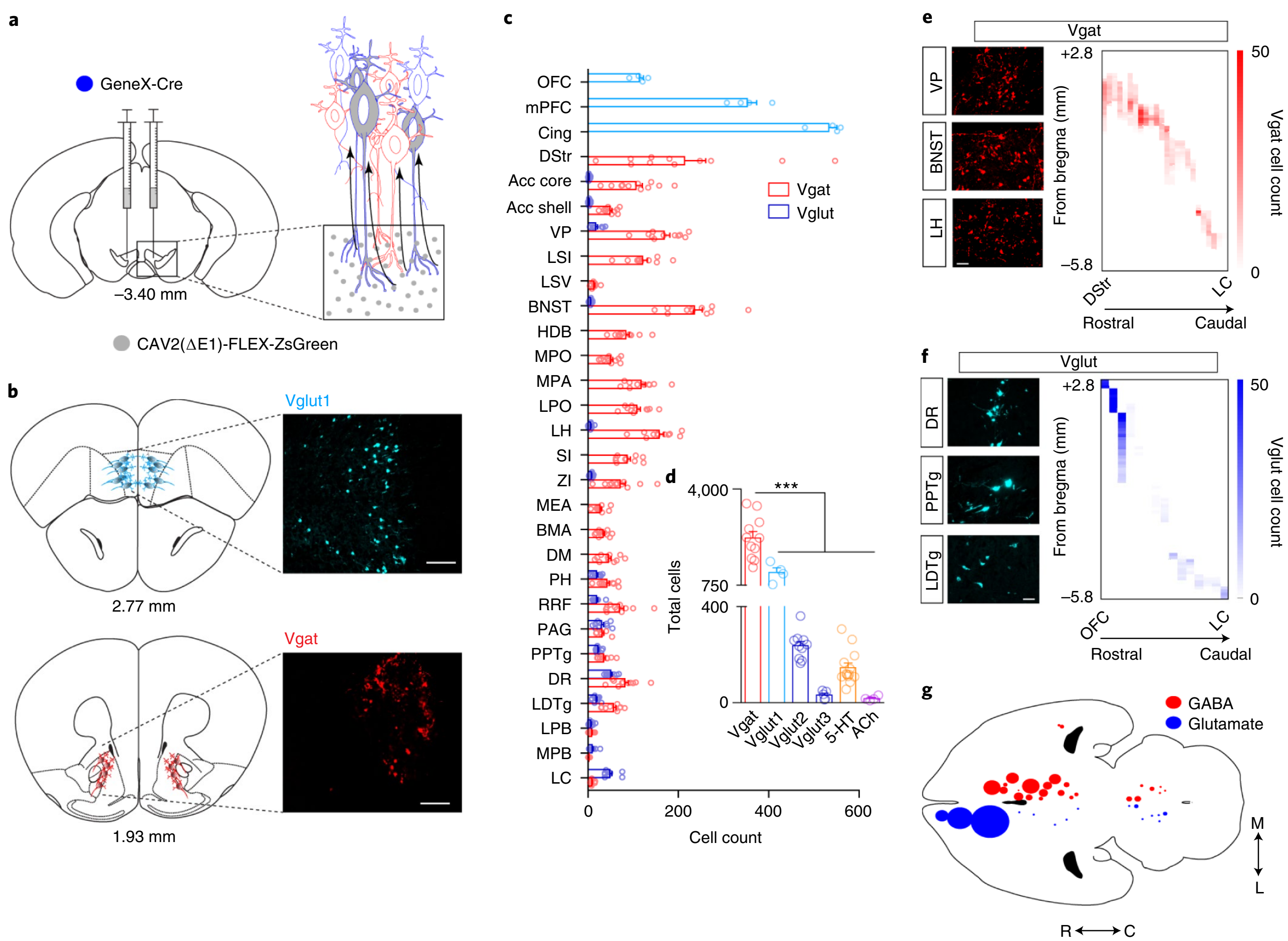


Fig. 1 | Retrograde mapping of neurotransmitter-specific inputs to the VTA. a, Illustration of CAV2-FLEX-ZsGreen injection into the VTA (gray circles) and retrograde transport. Note that all cells projecting to the VTA can take up the CAV2 virus, but only neurons of a designated cell type based on Cre expression can turn on ZsGreen expression (gray cells). **b**, Atlas illustrations and example images of ZsGreen-labeled cells (Vglut1-expressing, light blue; Vgat-expressing, red). Scale bar, 100 μm. **c**, Cell counts normalized to the total number of labeled neurons per mouse in identified brain regions from Vgat (red) or Vglut (blue) Cre-driver lines (Vgat, $n = 11$ mice; Vglut1, $n = 4$; Vglut2, $n = 10$; Vglut3, $n = 5$). BMA, basomedial amygdala; Cing, cingulate cortex; DM, dorsomedial hypothalamic nucleus; HDB/VDB, horizontal diagonal band/vertical diagonal band; LC, locus coeruleus; LPB, lateral parabrachial nucleus; LSI, lateral septum intermediate; LSV, lateral septum ventral; MEA, medial amygdala; MPA, medial preoptic area; MPB, medial parabrachial nucleus; PH, posterior hypothalamic area; RRF, retrorubral field; SI, substantia inominata; ZI, zona incerta. **d**, Total cell counts across the entire brain for each neurotransmitter phenotype (Vgat, $n = 11$ mice; Vglut1, $n = 4$; Vglut2, $n = 10$; Vglut3, $n = 5$; 5-HT, $n = 13$; ACh, $n = 4$; one-way ANOVA: $F_{(5,41)} = 60.62$, $P < 0.0001$, $^{***}P < 0.001$, Tukey's multiple comparisons test). **e**,**f**, Representative images of retrogradely labeled Vgat (**e**) and Vglut (**f**) cells in indicated regions, and heatmaps illustrating cell counts along the rostral (R)–caudal (C) axis (*y* axis) in each region (*x* axis). Scale bars, 50 μm. **g**, Cell density map illustrating relative number and location of projection neurons. Error bars represent the s.e.m.

inputs, we utilized the retrogradely transported canine adenoviral vector (CAV2) containing a conditional expression cassette for the bright fluorescent protein ZsGreen (CAV2-FLEX-ZsGreen)[24]. CAV2-FLEX-ZsGreen was injected into the VTA of six Cre-driver lines: Vgat-Cre (*Slc32a1*; GABA), Vglut1-, Vglut2- and Vglut3-Cre (*Slc17a7*, *Slc17a6* and *Slc17a8*; glutamate), ePet1-Cre (*Fev*; serotonin (5-HT)) and *Chat*-Cre (acetylcholine (ACh)) (Fig. 1a,b and Supplementary Table 1). Retrogradely labeled neurons were quantified every 90 μm throughout the rostral–caudal extent of the brain, based on anatomic registration according to an existing brain atlas[25] (Fig. 1b–d).

In total, we identified 29 brain regions that consistently displayed retrograde labeling (Fig. 1c–g and see also Extended Data Fig. 1). As expected, we identified one serotonergic input from the dorsal raphe nucleus (DRN) and two cholinergic inputs from the laterodorsal tegmental nucleus (LDTg) and the pedunculopontine tegmental nucleus (PPTg) (Fig. 1d and see also Extended Data Fig. 2). Glutamatergic inputs varied by region depending on the expression of Vglut1 (cortical), Vglut2 (subcortical) or Vglut3 (DRN), as expected (see Extended Data Fig. 1). GABAergic inputs derived from 26 of 29 total regions with the only exceptions in the cortex (Fig. 1b–e). Although glutamatergic inputs were derived from a variety of sources (16 of 29, Fig. 1b,c,f,g), GABAergic inputs outnumbered glutamatergic inputs in many regions (Fig. 1c and see also Extended Data Fig. 1).

GABAergic neurons were found across the rostral–caudal extent of the striatopallidal complex, the extended amygdala and the hypothalamus (Fig. 1c,e,g and see also Extended Data Fig. 1). In contrast, glutamatergic inputs were widespread throughout the orbitofrontal cortex (OFC), medial prefrontal cortex (mPFC) and cingulate cortex, and had substantial numbers of cells within the hindbrain but displayed relatively sparse labeling in other subcortical structures (Fig. 1c,f,g and see also Extended Data Fig. 1). We also observed numerous cells labeled in the VTA of Vglut2-Cre (511.8 ± 35.66 (mean ± s.e.m.)) and Vgat-Cre mice (713.2 ± 54.61) that displayed some differential rostral–caudal location within the VTA (see Extended Data Fig. 1). This observation is consistent with the reported local distribution and connectivity of these cell types[6,26,27], with previous estimations of more GABAergic than glutamatergic neurons within the VTA[14], and indicates the relative equivalency of viral transduction of Vglut2- and Vgat-expressing neurons by CAV2. Due to the locations of these neurons within the VTA, they were excluded from the total cell counts and analyses of projections to the VTA (Fig. 1a–g). We observed relatively few infected neurons in the substantia nigra (SN) pars compacta and pars reticulata adjacent to the VTA (see Extended Data Fig. 1), indicating that viral spread at the injection site was largely contained within the boundaries of the VTA.

We observed few Chat-positive neurons projecting to the VTA (Fig. 1d and see Extended Data Fig. 2) from the PPTg and LDTg, two brain regions shown to provide robust cholinergic modulation of the VTA[28]. It is possible that the CAV2 viral-based approach has differential tropism for these neurons. To address this question, we crossed the Chat-Cre mouse with the Ai14 TdTomato reporter line, and injected retrogradely transported fluorescent beads (RetroBeads) into the VTA. We observed many TdTomato$^+$ neurons in the PPTg and LDTg, and labeled numerous cells with RetroBeads in these regions, but very few Chat neurons expressing TdTomato co-labeled with RetroBeads (see Extended Data Fig. 2). Thus, although acetylcholine has a major influence on dopamine neuron activity, it appears to be derived from a relatively small number of neurons.

To further validate our retrograde CAV2 labeling, we injected RetroBeads into the VTA and performed RNAscope fluorescence in situ hybridization to label Vgat$^+$ and Vglut2$^+$ neurons in select regions that we observed have either biased GABAergic input over glutamatergic input (the bed nucleus of the stria terminalis (BNST) and lateral preoptic area (LPO)), or equivalent GABAergic and glutamatergic input (the periaqueductal gray (PAG)). Consistent with our CAV2 labeling, retrogradely labeled GABAergic neurons significantly outnumbered glutamatergic neurons in the BNST and LPO, and we found approximately equal numbers of retrogradely labeled GABAergic and glutamatergic neurons in the PAG (see Extended Data Fig. 3).

We found relatively few Vglut3$^+$ neurons projecting to the VTA from the DRN. Previous studies have shown a large number of Vglut3$^+$ neurons projecting from the DRN to the VTA[29] and a high degree of overlap between Vglut3 and the serotonergic marker TPH[29,30]. We found a significantly higher number of ePet1$^+$ serotonin neurons than Vglut3$^+$ neurons (see Extended Data Fig. 3), which is inconsistent with these previous reports. To investigate this further, we injected mice with RetroBeads and performed RNAscope in situ hybridization for Vgat and Vglut3, or Vgat and Vglut2. Consistent with our CAV2 mapping, we observed more retrogradely labeled Vgat$^+$ neurons than Vglut2$^+$ neurons in the DRN (see Extended Data Fig. 3); however, we observed significantly more Vglut3$^+$ neurons than Vgat$^+$ neurons, consistent with previous observations[29]. These findings indicate a more reliable identification of Vgat$^+$ and Vglut2$^+$ neurons using Vgat- and Vglut2-Cre relative to the Vglut3-Cre line. Both Vgat- and Vglut2-Cre drivers are IRES-Cre knock-in lines as opposed to Vglut3-Cre, which is a BAC transgenic. Whether the reduced efficiency is due to differences between mouse lines or the result of tropism is unclear. The efficacy of ePet1-Cre in identifying inputs to the VTA from the DRN and the high degree of Vglut3 coexpression in serotonin neurons makes it unlikely that tropism is the predominant, or only, cause of the observed discrepancy.

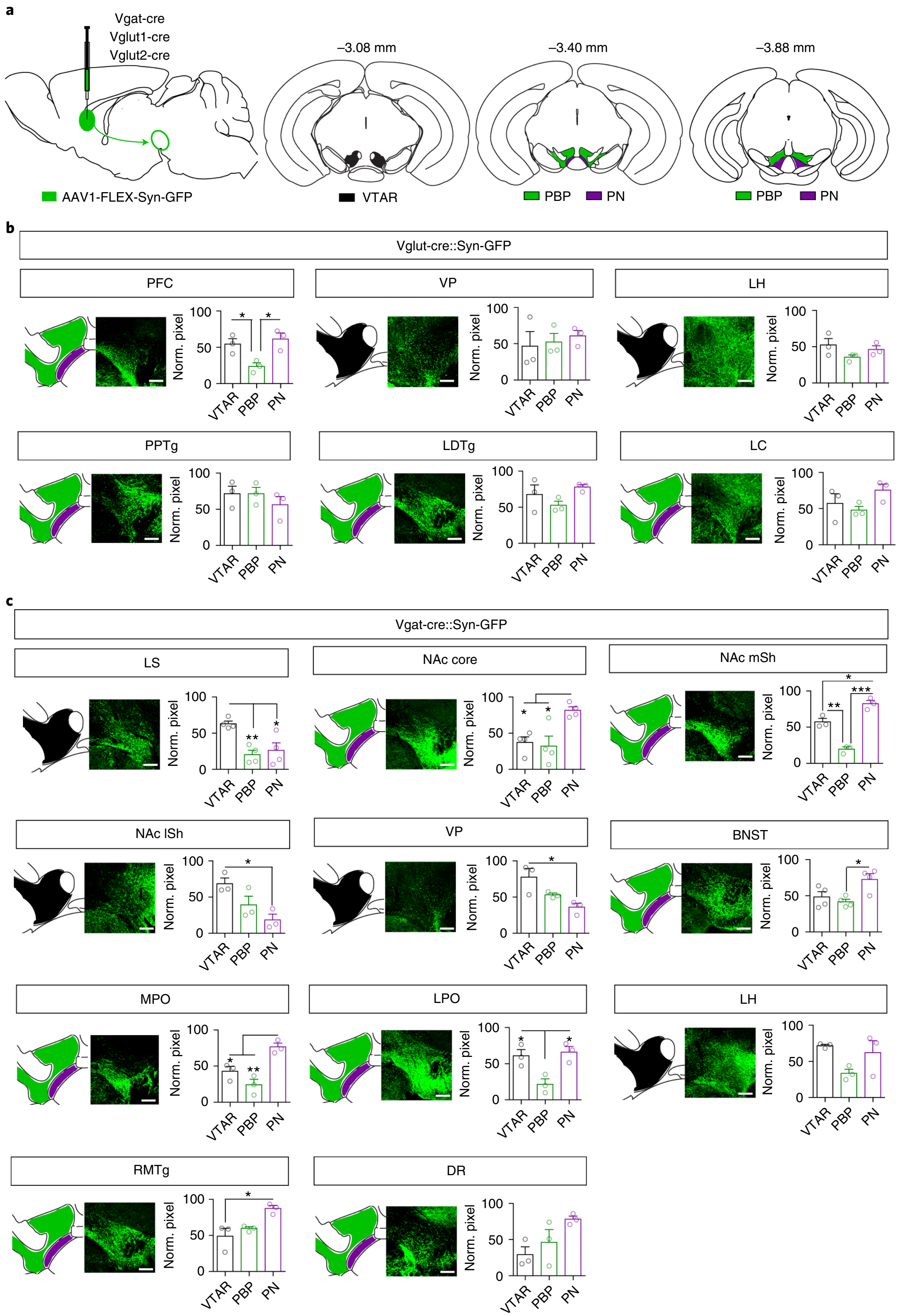
Differential glutamatergic and GABAergic innervation of the VTA. To confirm our retrograde analysis in Vgat-Cre, Vglut1-Cre and Vglut2-Cre mice, we injected approximately half the regions where projection neurons were identified (12 GABAergic and 6 glutamatergic) with the synaptic tracer AAV1-FLEX–synaptophysin-green fluorescent protein (Syn-GFP; Fig. 2 and see also Supplementary Figs. 1–3). Consistent with our retrograde tracing, we observed the greatest density of Syn-GFP puncta in the VTA from GABAergic inputs (see Extended Data Fig. 4). Analysis of CAV2-FLEX-ZsGreen cell number in a given region, versus Syn-GFP-integrated pixel density of VTA projections from that region, revealed a strong correlation (Pearson's $r = 0.7$, $P < 0.01$), with the exception of inputs from Vglut1::PFC (see Extended Data Fig. 4). For the dorsal striatum (DStr) we observed no measurable Syn-GFP projections in the VTA (see Extended Data Fig. 4); therefore, this structure was excluded from correlational analysis. For the PFC we did observe Syn-GFP puncta (Fig. 2), but to a lesser extent than predicted based on the large number of retrogradely labeled cells. Vgat::DStr and Vglut1::PFC send strong inputs to regions directly adjacent to the VTA: the substantia nigra pars reticulata (SNR) and the pontine nucleus, respectively (see Extended Data Fig. 4). This result highlights a caveat to this strategy where a small amount of viral spread outside the boundaries of the target region can lead to false-positive labeling. However, our projection analysis with Syn-GFP confirmed direct projections in 17 of 18 retrogradely labeled sites, validating the overall utility of this technique.

Analysis of Syn-GFP-labeled inputs revealed heterogeneity along the rostral–caudal axis of the VTA and within previously defined subdivisions (see Supplementary Figs. 4 and 5) particularly from GABAergic inputs. For better resolution of this heterogeneity, we compared the mean pixel intensities (normalized to peak) associated with three major subdivisions of the VTA (Fig. 2a): the VTAR, PN and PBP. Only one of six Vglut projections tested showed significant differential innervation of different subregions (PFC, Fig. 2b). By contrast, nine of eleven Vgat$^+$ projections showed statistically significant differential innervation (Fig. 2c).

Differential synaptic connectivity of glutamatergic and GABAergic inputs. To determine the synaptic connectivity patterns of inputs

Fig. 2 | Glutamatergic and GABAergic innervation of VTA subregions. a, Illustration of AAV1-FLEX–Syn-GFP injection into various Cre lines, and atlas images of the VTA showing designated VTAR, PBP and PN subdivisions, with indicated distance from bregma. **b**, Representative images and quantification of Syn-GFP projections in Vglut1 mice (PFC) or Vglut2 mice (all other regions) to the subdivisions of the VTA expressed as normalized (Norm.) pixel densities relative to the peak area of innervation ($n = 3$ mice per group; data measured from nine VTA sections per mouse (one-way ANOVA: PFC: $F_{(2,6)} = 8.34$, $P = 0.019$, $^*P < 0.05$, Tukey's multiple comparisons test). **c**, Representative images and quantification of Syn-GFP projections in Vgat mice to the subdivisions of the VTA expressed as normalized pixel densities relative to the peak area of innervation ($n = 3$ mice per group, 9 sections per mouse for all regions except $n = 4$ mice for LS, NAc core and BNST (one-way ANOVA: LS, $F_{(2,9)} = 10.08$, $P = 0.005$; NAc core, $F_{(2,9)} = 8.17$, $P = 0.01$; NAc mSh, $F_{(2,6)} = 57.87$, $P = 0.0001$; NAc lSh, $F_{(2,6)} = 7.02$, $P = 0.026$; VP, $F_{(2,6)} = 7.24$, $P = 0.025$; BNST, $F_{(2,9)} = 6.328$, $P = 0.019$; MPO, $F_{(2,6)} = 16.16$, $P = 0.004$; LPO: $F_{(2,6)} = 9.35$, $P = 0.014$; LH, $F_{(2,6)} = 4.162$, $P = 0.085$; RMTg, $F_{(2,6)} = 8.34$, $P = 0.019$; DR: $F_{(2,9)} = 4.162$, $P = 0.074$; $^*P < 0.05$, $^{**}P < 0.01$, $^{***}P < 0.001$, Tukey's multiple comparisons test). Error bars represent s.e.m. Scale bar, 200 μm.

to the VTA, we targeted expression of channelrhodopsin (ChR2) to eleven regions that displayed the highest level of input or most regionally restricted innervation (four glutamatergic and seven GABAergic). To target glutamatergic neurons, Vglut1- or Vglut2-Cre mice were crossed with mice expressing Flp recombinase under the control of the tyrosine hydroxylase (*Th*) promoter



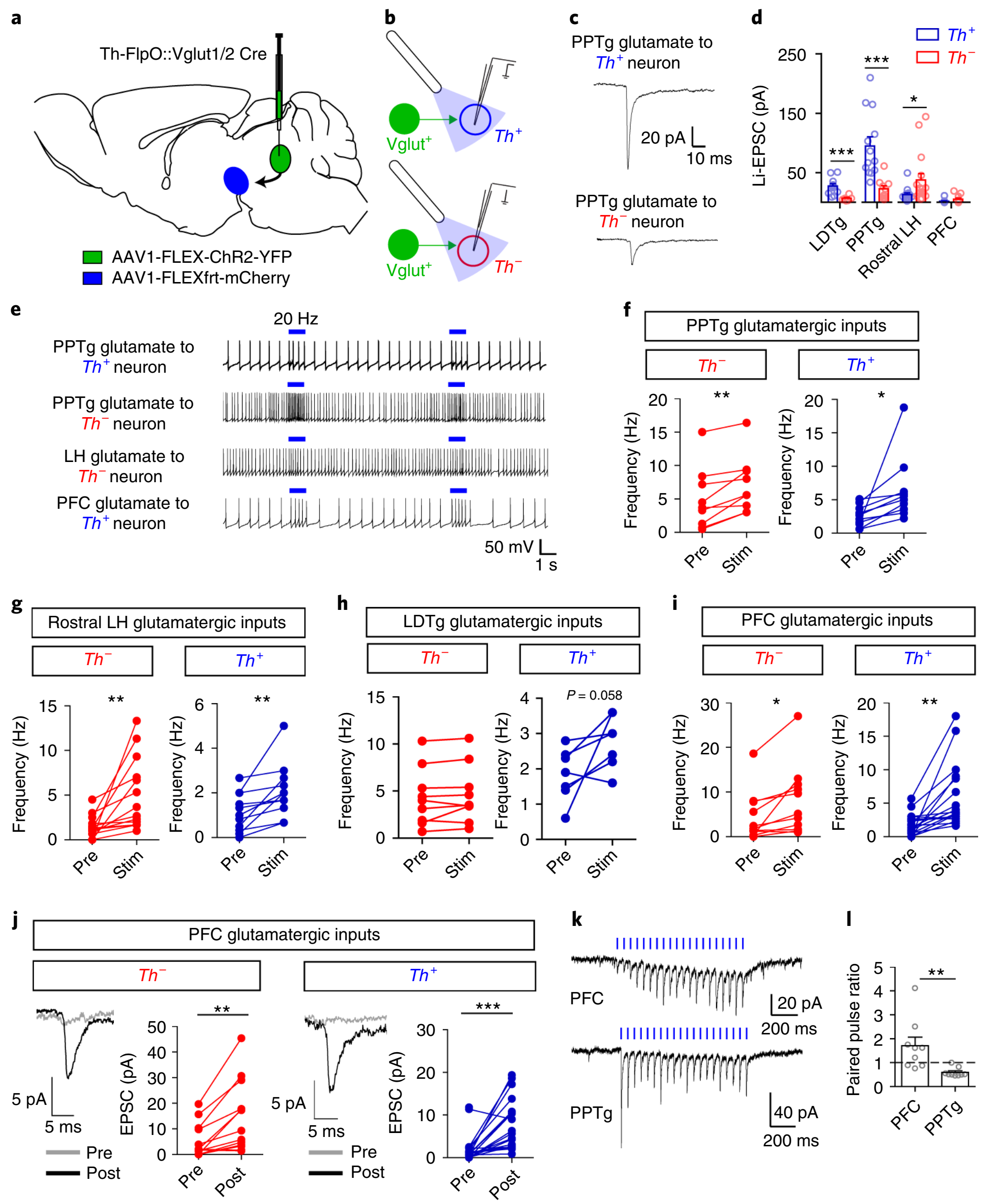


Fig. 3 | Connectivity of glutamatergic inputs to the VTA. **a**, Illustration of AAV1-FLEX-ChR2-YFP injection into the region of interest and AAV1-FLEXfrt-mCherry injection into the VTA in Th-FlpO::Vglut Cre mice. **b**, Illustration of patch-clamp recordings from *Th*⁺ (dopamine) and *Th*⁻ (nondopamine) VTA neurons and blue light stimulation of ChR2 inputs. **c**, Example traces (average of 10 sweeps). **d**, Quantification of Li-EPSCs recorded from *Th*⁺ and *Th*⁻ neurons in the VTA, identified by fluorescence. Only cells with a measurable connection are included here; for PFC, the amplitudes are measured before 20-Hz stimulation (see Extended Data Fig. 5 for numbers of connected versus unconnected cells for each input; LDTg, *Th*⁺, $n = 11$ cells; *Th*⁻, $n = 6$; PPTg, *Th*⁺, $n = 13$; *Th*⁻, $n = 10$; LH: *Th*⁺, $n = 14$; *Th*⁻, $n = 16$; PFC: *Th*⁺, $n = 23$; *Th*⁻, $n = 18$; Student's two-tailed *t*-test: LDTg, $P = 0.004$; PPTg, $P = 0.001$; LH, $P = 0.045$.) **e**, Example traces of action potential firing; blue bar indicates 1 s of 20-Hz light stimulus. **f–i**, Quantification of action potential frequency before (Pre) and during (Stim) 20-Hz light stimulation of inputs from the PPTg (**f**), LH (**g**), LDTg (**h**) or PFC (**i**), averaged from three sweeps per cell (PPTg: *Th*⁻, $n = 9$ cells; *Th*⁺, $n = 10$; LDTg: *Th*⁻, $n = 9$; *Th*⁺, $n = 7$; LH: *Th*⁻, $n = 14$; *Th*⁺, $n = 12$; PFC: *Th*⁻, $n = 11$; *Th*⁺, $n = 18$; paired, two-tailed Student's *t*-tests; PPTg *Th*⁻, $P = 0.003$; PPTg *Th*⁺, $P = 0.034$; LH *Th*⁻, $P = 0.005$; LH *Th*⁺, $P = 0.005$; PFC *Th*⁻, $P = 0.011$; PFC *Th*⁺, $P = 0.001$). **j**, Example traces (average of 10 sweeps) and Li-EPSC amplitudes recorded before and after five rounds of 1 s of 20-Hz light stimulation (*Th*⁻, $n = 13$ cells, *Th*⁺, $n = 23$ cells; paired, two-tailed Student's *t*-tests: *Th*⁻, $P = 0.002$; *Th*⁺, $P = 0.0002$). **k**, Example traces of Li-EPSCs from PFC and PPTg inputs evoked by 20-Hz light stimulus. **l**, Paired pulse ratio (second pulse amplitude:first pulse amplitude) of Li-EPSCs from PFC and PPTg inputs ($n = 9$ cells per group; Student's two-tailed *t*-test, $P = 0.007$). For all panels, error bars represent s.e.m.

(a dopamine neuron marker)[31]. AAV1-FLEX-ChR2-YFP (where YFP is yellow fluorescent protein) was injected into the region of interest, and the Flp-dependent virus AAV1-FLEXfrt-mCherry was injected into the VTA to label dopamine neurons (Fig. 3a,b). Analysis of light-evoked excitatory postsynaptic currents (Li-EPSCs) revealed that Vglut2 inputs to the VTA from the LDTg and PPTg showed a significant connectivity bias to dopamine neurons over nondopamine neurons, with PPTg inputs being the strongest (Fig. 3c,d and see also Extended Data Fig. 5). By contrast, Vglut2 inputs from the rostral lateral hypothalamic area (LH) to the VTA exhibited a significant connectivity bias to nondopamine neurons, similar to previous reports[20]. Surprisingly, Li-EPSCs from the

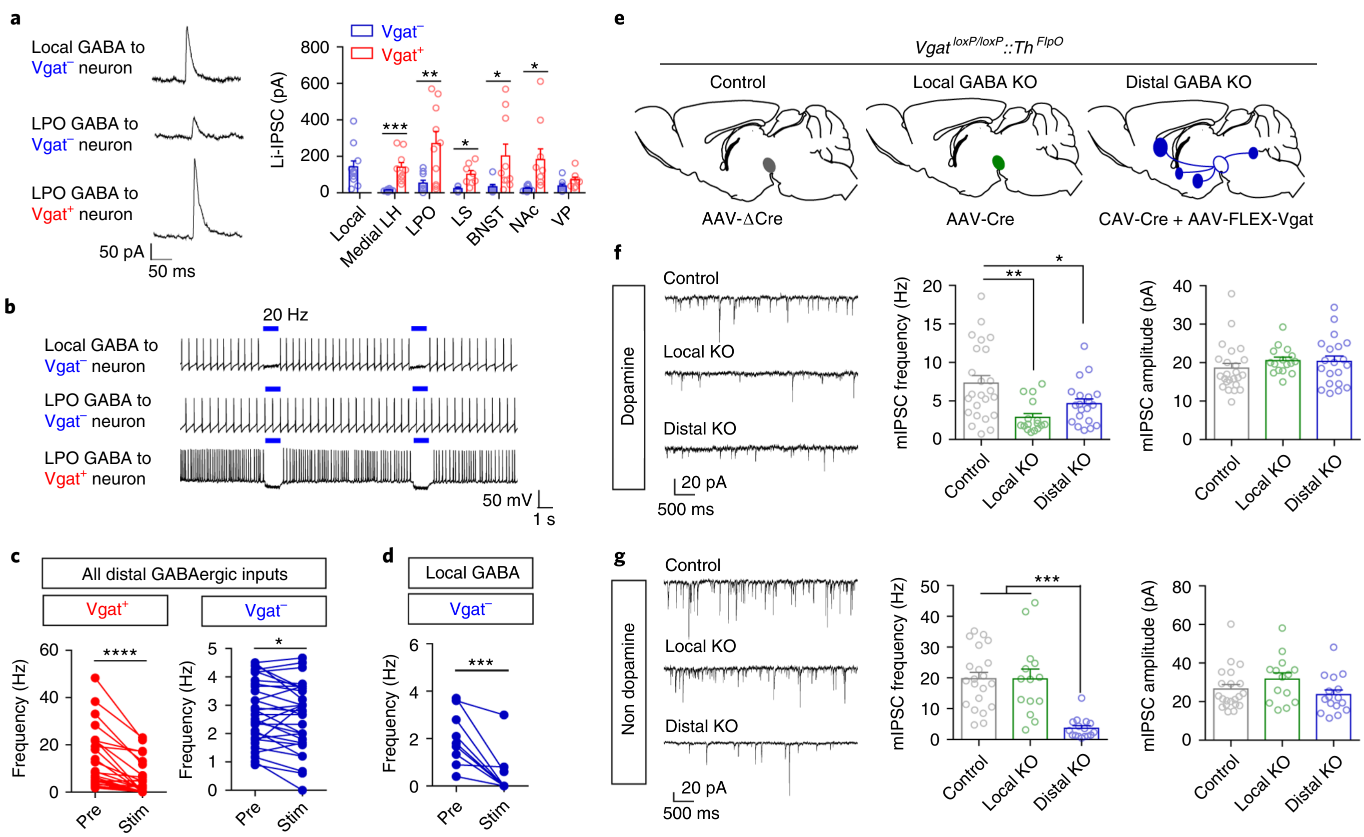


Fig. 4 | Connectivity of GABAergic inputs to the VTA. a, Example traces (average of 10 sweeps) and quantification of Li-IPSCs recorded from Vgat⁻ and Vgat⁺ neurons in the VTA, identified by fluorescence. Only cells with a measurable connection are included here (see Extended Data Fig. 5 for numbers of connected versus unconnected cells for each input; local $n=11$ cells; LH: Vgat⁻, $n=8$; Vgat⁺, $n=10$; LPO: Vgat⁻, $n=10$, Vgat⁺, $n=10$; LS: Vgat⁻, $n=5$; Vgat⁺, $n=8$; BNST: Vgat⁻, $n=8$, Vgat⁺, $n=10$, NAc: Vgat⁻, $n=10$; Vgat⁺, $n=10$; VP: Vgat⁻, $n=9$, Vgat⁺, $n=8$; Student's two-tailed t-test: LH, $P=0.0004$; LPO, $P=0.005$; LS, $P=0.011$; BNST, $P=0.036$; NAc, $P=0.016$). **b**, Example traces of action potential firing; blue bar indicates 1 s of 20-Hz light stimulus. **c**,**d**, Quantification of action potential frequency before (Pre) and during (Stim) 20-Hz light stimulation of distal GABA inputs (**c**) or local GABA neurons (**d**), averaged from three sweeps per cell (distal Vgat⁺, $n=28$ cells; distal Vgat⁻, $n=33$; local $n=10$; paired, two-tailed Student's t-tests: distal Vgat⁺, $P<0.0001$; distal Vgat⁻, $P=0.017$; local Vgat⁻, $P=0.0007$). **e**, Illustration of experimental groups. Vgat$^{loxP/loxP}$::*Th*^{FlpO} mice were injected in the VTA with the indicated virus(es) mixed with AAV1-FLEXfrt-mCherry to label dopamine neurons for slice recording. **f**,**g**, Example traces and quantification of mIPSC frequency and amplitude recorded from *Th*⁺ (**f**) and *Th*⁻ (**g**) neurons (*Th*⁺: $n=24$ cells control, 16 VTA KO, 20 distal KO; *Th*⁻: 21 cells control, 14 cells VTA KO, 16 cells distal KO; one-way ANOVA: *Th*⁺ frequency $F_{(2,57)}=7.48$, $P=0.001$; *Th*⁻ frequency $F_{(2,48)}=17.31$, $P<0.0001$; $^*P<0.05$, $^{**}P<0.01$, $^{***}P<0.001$, Tukey's multiple comparisons test). Error bars represent s.e.m.

mPFC inputs were almost undetectable (<5 pA) in most dopamine and nondopamine neurons (Fig. 3d).

To determine the efficacy of specific inputs in regulating dopamine neuron firing, action potentials were recorded in response to 20-Hz light stimulation. Although we observed a projection bias of glutamatergic inputs from the PPTg and rostral LH, stimulation of these terminals significantly increased action potential firing in both dopamine and nondopamine neurons (Fig. 3e–g). Stimulation of LDTg inputs, which were weaker than those from the PPTg, did not significantly increase firing in nondopamine neurons and was not reliable at evoking increased action potential firing in dopamine neurons (Fig. 3h).

To our surprise, mPFC inputs that did not reliably evoke detectable Li-EPSCs drove significant increases in action potential firing in both dopamine and nondopamine populations (Fig. 3i). Consistent with this observation, we detected reliably evoked Li-EPSCs in dopamine and nondopamine neurons after the 20-Hz stimulus trains (Fig. 3j). Recording Li-EPSCs during the stimulus train showed a strongly facilitating synapse (paired pulse ratio >1), in contrast to inputs from the PPTg, which were depressing (Fig. 3k,l). Stimulus trains of 10 Hz, but not 5 Hz, were sufficient to induce this significant increase in amplitude (see Extended Data Fig. 5). We found that the amplitude of enhanced PFC Li-EPSCs was maintained for at least 30 min after stimulation; by contrast, repetitive stimulation of PPTg inputs did not cause a significant increase in response amplitude (see Extended Data Fig. 5). The enhancement and maintenance of Li-EPSC amplitude were not blocked in the presence of the *N*-methyl-D-aspartate (NMDA)-receptor antagonist AP5 (see Extended Data Fig. 5), suggesting that it is not an NMDA-receptor-dependent mechanism.

To establish the synaptic connectivity of GABA inputs to the VTA, Vgat-Cre mice were injected with AAV1-FLEX-ChR2-mCherry into the region of interest and AAV1-FLEX-YFP into the VTA to label GABA neurons for patching. Analysis of light-evoked inhibitory postsynaptic currents (Li-IPSCs) revealed robust connectivity from local VTA GABA neurons to non-GABA VTA neurons (Fig. 4a). In contrast, all GABAergic inputs from outside the VTA showed a preferential connectivity bias toward VTA GABA neurons compared with non-GABA neurons, with the exception of the ventral pallidum (VP), which showed no significant bias (Fig. 4a and see also Extended Data Fig. 5), consistent with previous observations[22].

We next recorded changes in action potential firing during 20-Hz stimulation (Fig. 4b). Due to the relative homogeneity of synaptic connectivity of distal GABAergic inputs, we pooled these samples for simplicity. Consistent with the connectivity bias of these inputs, we observed significant suppression of action potential firing in VTA GABA neurons, but much weaker suppression in non-GABA neurons (Fig. 4c). In contrast, local GABA neurons potently suppressed the action potential firing of non-GABA neurons (Fig. 4d).

These results suggest that local GABA-producing neurons within the VTA/RMTg predominantly synapse on to non-GABA-producing neurons, whereas descending GABAergic inputs predominantly synapse on to local GABAergic neurons within the VTA. We were unable to assess the degree to which local GABAergic neurons synapse on to other GABA-releasing neurons within the region because of ChR2 expression within neighboring cells. For better resolution of this we devised a genetic strategy to selectively inactivate GABA release from local or distal GABA sources (Fig. 4e). To achieve this, we crossed a mouse line in which the gene encoding Vgat (*Slc32a1*) is flanked by *loxP* sites with the Th^{FlpO} mouse line to generate $Vgat^{loxP/loxP}$::$Th^{FlpO/+}$ mice. Control mice were injected in the VTA with a nonfunctional Cre virus (AAV1-ΔCre-GFP), along with AAV1-FLEXfrt-mCherry, to label dopamine neurons. To inactivate local sources of GABA release, we injected AAV1-Cre-GFP and AAV1-FLEXfrt-mCherry into $Vgat^{loxP/loxP}$::$Th^{FlpO/+}$ mice. To inactivate distal GABA release, we injected CAV2-Cre along with AAV1-FLEXfrt-mCherry and AAV1-FLEX-Vgat into $Vgat^{loxP/loxP}$::$Th^{FlpO/+}$ mice. As CAV2 does infect neurons at the site of injection (see Extended Data Fig. 1), we included a 'rescue' virus (AAV1-FLEX-Vgat) to restore Vgat expression to neurons within the VTA that were infected with CAV2-Cre[32].

To validate the effectiveness of these viral strategies we utilized RNAscope in situ hybridization to label Cre^+ and $Vgat^+$ neurons in the VTA. AAV1-Cre-GFP reduced the number of Vgat-expressing neurons in the VTA to 18.4% of control, with 2.1% of Cre^+ cells expressing Vgat. By contrast, injection of CAV2-Cre^+ AAV1-FLEX-Vgat restored the number of $Vgat^+$ neurons above control levels, with 99.4% of Cre^+ cells expressing Vgat (see Extended Data Fig. 6).

To establish the impact of inactivating GABA release from local versus distal sources, we patched Th^+ and Th^- neurons within the VTA and recorded mIPSCs. Inactivation of Vgat within the VTA reduced mIPSCs on Th^+ neurons by approximately 60% (Fig. 4f), but had no significant effect on mIPSCs recorded from Th^- neurons (Fig. 4g), suggesting little connectivity between local GABAergic neurons and nondopamine-producing neurons within the region. Inactivation of Vgat from distal sources significantly reduced synaptic GABA release on to Th^+ neurons (Fig. 4f), suggesting that the small Li-IPSCs, recorded from all descending inputs to non-GABA neurons (Fig. 4a), sum to approximately 40% of all synaptic inhibitory contact on to dopamine-producing cells. Consistent with the strong bias of Li-IPSCs on to GABAergic neurons of the VTA, inactivation of distal GABA release reduced mIPSCs by over 80% on to Th^- neurons (Fig. 4g). Neither manipulation affected mIPSC amplitude (Fig. 4f,g). These data indicate that dopamine neurons receive significant GABAergic input from both local and distal sources, whereas GABAergic input on to VTA GABA neurons arises almost entirely from distal sources.

Distal and local GABA inputs to the VTA differentially regulate behavior. Our ability to genetically inactivate GABA release from local or distal sources gave us the unique opportunity to investigate how these sources impact behavior. We previously demonstrated that disruption of local GABAergic networks within the VTA profoundly alters psychomotor behavior[18]. Consistent with this observation, inactivation of local GABA release resulted in high levels of basal locomotor activity, whereas inactivation of release from distal GABA sources had little effect on this behavior (see Extended Data Fig. 6).

GABA in the VTA is increasingly linked to the regulation of the psychomotor response to cocaine[4,18,33,34]. To determine whether local or distal GABA differentially impacts psychomotor sensitization to cocaine, mice were injected (20 mg kg^{-1}, subcutaneously) once daily for 5 consecutive days (Fig. 5a). Cocaine elicited a robust paradoxical calming of hyperlocomotor activity in local GABA knockout (KO) mice (Fig. 5b,c). In contrast, cocaine injection resulted in increased locomotor response in control and distal GABA KO mice, which increased across days (Fig. 5b,c). This effect was potentiated in distal GABA KO mice (Fig. 5b,c).

To determine whether local or distal sources of GABA to the VTA differentially impact basic reward-related behavior, we tested mice in a simple Pavlovian conditioning paradigm in which presentation of an auditory and visual cue (lever extension) co-terminated with delivery of a sucrose pellet reward (Fig. 5d). Control mice demonstrate a learned association between the cue and reward by making increased anticipatory head entries into the food hopper during the cue presentation (Fig. 5e). In contrast, local GABA KO mice failed to demonstrate a conditioned reward association (Fig. 5f). Distal GABA KO mice learned this association at a similar rate to controls (Fig. 5g). Although local GABA KO mice are hyperactive, their total head entries during each session did not differ from controls or distal GABA KO mice (see Extended Data Fig. 6).

After Pavlovian conditioning, mice were switched to an operant paradigm in which reward delivery (one sucrose pellet) was contingent on making a single lever press (FR1) on either of the extended levers (Fig. 5h). Over 3 d of training there was no difference in total lever pressing between control and the distal GABA KO mice (see Extended Data Fig. 6), but local GABA KO mice pressed at a significantly higher rate (see Extended Data Fig. 6).

Next, we asked whether mice could distinguish between a high- or low-value reward. If given a choice between two levers in an operant task, most mice will display a strong and consistent preference (~75%) for one lever or the other. We observed a strong lever bias in all three groups of mice (see Extended Data Fig. 6). After 3 d of operant conditioning, mice were subjected to a lever-switching task in which a press on their preferred lever continued to deliver a single pellet, whereas a press on their nonpreferred lever delivered three pellets. After 5 d of conditioning, control mice switched their preference to the high-reward lever (Fig. 5i). Local GABA KO mice maintained a persistent preference for the originally preferred (low-reward) lever (Fig. 5j). Likewise, distal GABA KO mice did not switch their preference, although they did not persevere as long as local GABA KO mice (Fig. 5k).

In addition to Pavlovian reward association, the VTA is also implicated in discriminatory fear learning[35]. To assess whether Pavlovian threat discrimination is impacted by local or distal GABA KO, mice were assessed in a discriminatory threat-conditioning paradigm[35] in which a predictive conditioned stimulus (CS^+) tone was paired ten times with unconditioned stimulus (US; 0.3 mA foot shock) delivery. Ten unpaired CS^- tone presentations of a different frequency were randomly interleaved with CS^+–US pairings. After 2 d of conditioning, both control and distal GABA KO mice could discriminate between the CS^+ and CS^-, as measured by time spent freezing during cue presentation. Local GABA KO mice did not discriminate between CS^+ and CS^-, displaying similar freezing response to both cues (see Extended Data Fig. 6).

In addition to the regulation of reward learning, dopamine plays an important role in regulating social behaviors[36] and descending disinhibitory inputs from the medial preoptic nucleus (MPO) have been shown to play an important role in this process[21]. To determine whether local or distal GABA KO differentially impacts social behavior, we tested mice in a three-chamber task to measure social preference and social recognition. After a period of habituation to a three-chambered arena, an unfamiliar mouse contained within a small wire cage was placed in one chamber and an empty cage (object) was placed in the opposite chamber (Fig. 5l). Exploratory behavior was measured for 5 min. Local GABA KO mice showed the highest level of preference for the novel mouse over the novel object (Fig. 5m,n). Distal GABA KO mice showed no preference (Fig. 5m,n). Next, mice were allowed to habituate to the unfamiliar mouse for a period of 30 min, after which a novel mouse was placed in the chamber in the empty cage (Fig. 5l). Control mice and local GABA KO mice showed a significant preference for the novel mouse over the familiar mouse, whereas distal KO mice did not show a preference for either mouse (Fig. 5o–p).

Individual GABAergic inputs differentially regulate dopamine-dependent behaviors. Our projection mapping analysis revealed that distal GABA inputs display differential innervation of VTA subdivisions. To determine whether a subset of these inputs differentially influences reinforcement behavior, we conditionally expressed ChR2 in four brain regions with different innervation patterns: the lateral septum (LS), which predominantly innervates the VTAR, the medial nucleus accumbens shell (NAc mSh) and the MPO, which both predominantly innervate the PN but have subtly different innervation of the PBP, and the LH, which had the most broadly distributed innervation pattern (see Fig. 2).

Vgat-Cre mice were injected with AAV1-FLEX-YFP (control) or AAV1-FLEX-ChR2-YFP in the indicated regions and fiberoptic cannulas were implanted above the VTA (Fig. 6a and see also Supplementary Figs. 6 and 7). Blue light stimulation (20 Hz, 5-ms pulses for 20 min) led to an increase in cFos$^+$ dopamine neurons in VTA subregions with strong ChR2-fiber innervation (Fig. 6b and see also Extended Data Fig. 7). Only mice injected with ChR2 in the LH showed a significant increase in cFos in the PBP, whereas mice injected in the LH, the NAc mSh and the MPO all showed an increase in cFos in the PN. We did not observe a significant increase in cFos staining in mice injected in the LS. Intriguingly, in LH-, NAc- and MPO-injected mice the region with the highest number of cFos$^+$ neurons was the small IF nucleus, located between the two halves of the paranigral subdivision (Fig. 6b and see also Extended Data Fig. 7).

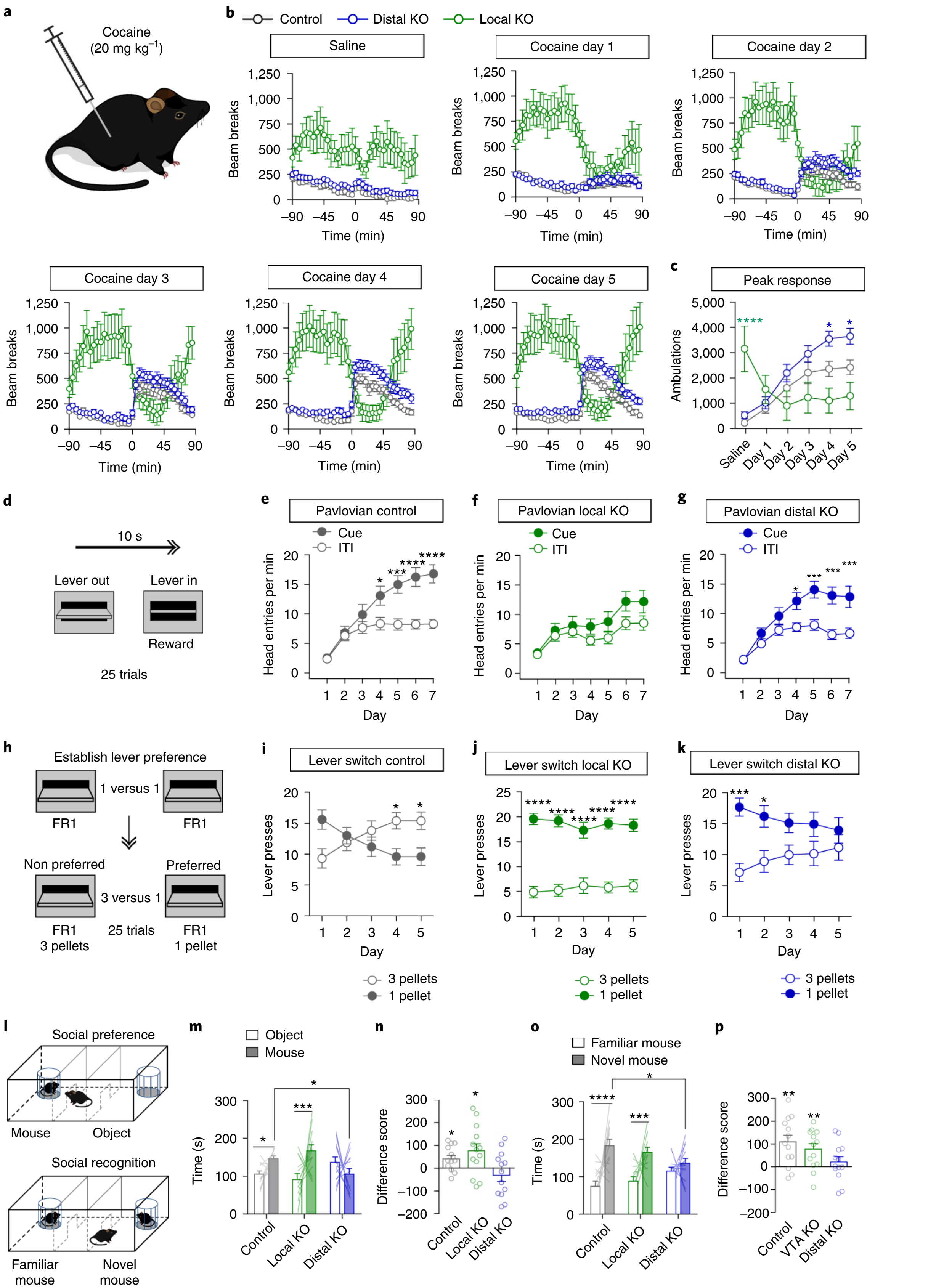
To directly compare behavioral outcomes associated with activation of these inputs, we first tested mice in a real-time place preference (RTPP) assay. Blue light stimulation (20 Hz) of LH or MPO GABA inputs, but not LS or mSh inputs, on one side of a two-chambered arena caused a robust place preference (Fig. 6c,d). VTA stimulation of mSh inputs caused a significant increase in shuttles between chambers (Fig. 6e) which was associated with an increase in distance traveled (Fig. 6f). MPO mice did not show a significant increase in shuttles (Fig. 6e), but did show increased locomotor activity (Fig. 6f). In contrast, LH mice showed a significant decrease in locomotor behavior (Fig. 6f). We observed that LH mice spent most of their time in the light-paired side attempting to gnaw at the walls or the doorway of the arena, similar to previous observations[20,37], which accounts for their reduced locomotion. To quantify this behavior across groups, we placed each mouse in an empty cage with a single standard food pellet and delivered 20-Hz light stimulation for 20 min. Stimulation of LH inputs resulted in a robust gnawing of the food pellet and accumulation of pellet dust on the cage floor, as evidenced by the significant reduction in the weight of the pellet (Fig. 6g). This effect was not observed with stimulation of other inputs to the VTA.

To probe the extent to which activation of these inputs is reinforcing further, mice were assayed in a simple fixed ratio schedule of optical self-stimulation. Mice were given 1 h of training per day for 5 d, in which a single lever press on either extended lever resulted in 3 s of 20-Hz light stimulation. LH mice exhibited robust lever-pressing behavior, whereas MPO and NAc mice exhibited moderate lever pressing that eventually reached statistical significance relative to YFP control mice (Fig. 6h,i). For the three groups of mice that showed significant self-stimulation, we next switched their preferred lever from 20-Hz stimulation to 7-Hz stimulation, similar to the three pellet versus one pellet lever-switching task (Fig. 5h). All three groups of mice rapidly switched to their nonpreferred lever to continue receiving 20-Hz stimulation, with the most rapid switching observed in LH mice (Fig. 6j–l).

Inactivation of distal GABA inputs to the VTA disrupts social behavior in the three-chamber assay. To assess which if any of the four inputs to the VTA influence this behavior, we assessed mice with optical inhibition of the inputs using the inhibitory opsin Jaws[38]. Vgat-Cre mice were injected with AAV1-FLEX-Jaws-GFP or AAV1-FLEX-YFP (control) into the same four input regions and optic fibers were implanted over the VTA (Fig. 6m and see also Supplementary Figs. 6 and 7). Red light activation of Jaws (2 s on, 1 s ramp down, 2 s off[35]) in one chamber of a two-chamber arena did not cause a significant preference or aversion in any group (see Extended Data Fig. 7). During the three-chamber social preference assay, mice received red light whenever they were in the chamber containing the target mouse (Fig. 6n). Inhibition of mSh and MPO

Fig. 5 | Vgat knockout from local and distal sources differentially affects reward behaviors. a, Mice received saline or cocaine injections (subcutaneously, 20 mg kg^{-1}). **b**, Locomotor activity, measured as infrared beam breaks, before and after saline or cocaine injection at time zero (control, $n=16$ mice; distal KO, $n=18$; VTA KO, $n=10$). **c**, Peak locomotor response, summed from 15 min to 45 min post-injection (control, $n=16$ mice; distal KO, $n=18$; VTA KO, $n=10$; two-way repeated measures (RM) ANOVA: distal KO versus control, effect of virus $F_{(1,32)}=6.05$, $P=0.02$; VTA KO versus control, interaction $F_{(5, 115)}=13.58$, $P<0.0001$; Bonferroni's multiple comparisons: $^*P<0.05$, $^{****}P<0.0001$.) **d**, Schematic of Pavlovian conditioning with lever extension for 10 s serving as a compound visual–auditory cue, and lever retraction coinciding with reward delivery. **e–g**, Average head entries per minute during the cue or the ITI period on each day of Pavlovian conditioning in control (**e**), local KO (**f**) or distal KO (**g**) mice ($n=23$ mice per group control: 21 VTA KO, 27 distal KO; two-way RM ANOVA: control Interaction $F_{(6, 264)}=7.16$, $P<0.0001$, distal KO Interaction $F_{(6, 312)}=4.21$, $P=0.0004$; Bonferroni's multiple comparisons: $^*P<0.05$, $^{***}P<0.001$, $^{****}P<0.0001$). **h**, Schematic of FR1 conditioning (3 d) followed by lever-switching paradigm in which the value of the nonpreferred lever increases to three pellets, whereas the value of the preferred lever remains the same. **i–k**, Presses on high- and low-reward levers over 5 d of training in control (**i**), local KO (**j**) or distal KO (**k**) mice ($n=15$ mice per group control, 11 VTA KO, 15 distal KO; two-way RM ANOVA: control Interaction $F_{(4, 112)}=10.96$, $P<0.0001$; VTA KO effect of lever $F_{(1,20)}=73.48$, $P<0.0001$, distal KO Interaction $F_{(4, 112)}=5.40$, $P=0.0005$; Bonferroni's multiple comparisons: $^*P<0.05$, $^{***}P<0.001$, $^{****}P<0.0001$). **l**, Schematic of social preference and social recognition assay. **m**, Time spent in the mouse or object chamber during the 5-min trial period ($n=13$ mice per group; two-way ANOVA interaction $F_{(2,36)}=4.56$, $P=0.017$; mouse chamber: distal KO versus control $^*P<0.05$; Bonferroni's selected comparisons between subjects: mouse versus object: control $^*P<0.05$; local KO $^{***}P<0.001$, Fisher's two-tailed single within-subject comparison). **n**, Difference score (time in mouse chamber – time in object chamber; two-tailed, one-sample t-test, theoretical mean = 0; control $P=0.018$, local KO $P=0.031$). **o**, Time spent in the familiar or novel mouse chamber during the 5-min trial period ($n=13$ mice per group; two-way ANOVA Interaction $F_{(2,36)}=5.60$, $P=0.006$; mouse chamber: distal KO versus control $^*P<0.05$; Bonferroni's selected comparisons between subjects: novel mouse versus familiar mouse: control $^{****}P<0.0001$; local KO $^{***}P<0.001$, Fisher's two-tailed single within-subject comparison). **p**, Difference score (time in novel mouse chamber – time in familiar mouse chamber; two-tailed, one-sample Student's t-test, theoretical mean = 0, control: $P=0.003$; local KO: $P=0.007$). Error bars represent s.e.m.

inputs disrupted social preference that was not observed in the other groups (Fig. 6o). Next, we probed whether inhibition during social preference would disrupt social recognition, when inputs were no longer actively inhibited. Inhibition of MPO and mSh inputs during social preference testing impaired social recognition that was not observed in the other groups (Fig. 6p).



Discussion

The VTA receives synaptic input from a large number of brain regions that converge on this critical structure and enable the complex encoding of information in the form of precise patterns of dopamine neuron firing and downstream dopamine release[2,5,8,10,12,13,39,40]. Although previous studies have identified inputs to specific cell types and/or subregions of the VTA[8,10,12,13], in the present study we provide a large-scale map of the neurotransmitter identity of these VTA afferents. From these findings we have gained a new understanding of how inputs to the VTA are organized, with the strongest glutamatergic inputs originating from descending cortical projections and ascending hindbrain projections, whereas inputs from most other subcortical structures are largely GABAergic. Moreover, we find that, although glutamatergic inputs, with the exception of those coming from the mPFC, tend to innervate all subregions of the VTA with similar intensity, most GABAergic inputs exhibited biased innervation toward specific VTA subregions.

One potential caveat of our approach is that CAV2 may display tropism, or uneven infection of different cell types. As we identified only a small number of VTA-projecting cholinergic cells in the LDTg and PPTg with our CAV strategy, we suspected that perhaps these neurons are not well labeled by this virus. However, we identified a similarly small number of Chat$^+$ cells labeled by RetroBeads, indicating that this result is probably accurate, and that a small number of cholinergic neurons in this region are responsible for the VTA projection. This does not discount the importance of cholinergic regulation of the VTA. Indeed, cholinergic inputs to the VTA regulate reward reinforcement[28], and potently influence dopamine neuron activity patterns[41]. It is also possible that CAV displays varying tropism for either GABA or glutamatergic neurons. However, combining RetroBead labeling with in situ hybridization for Vgat and Vglut2 in select regions confirmed the overall pattern of results we saw with CAV labeling. Our results are largely consistent with a previous analysis of glutamatergic inputs to the VTA in rats using wheat germ agglutinin retrograde tracing[15]. Although these investigators found different contributions of glutamatergic inputs to the VTA, such as a greater overall proportion of glutamatergic projections from the LH relative to hindbrain inputs, they also observed that most retrogradely labeled neurons in subcortical regions were not glutamatergic. The exact reason for these observed differences is not clear, but may reflect differences in the species investigated (rat versus mouse), the injection site location or the retrograde tracing method.

We observed that many of the GABAergic inputs to the VTA synapse strongly on to GABA neurons and more weakly on to non-GABA neurons. Previous studies examining individual regions including the NAc, BNST and LH have seen similar results[11,19–23]. The VTA also contains a significant population of glutamatergic neurons[1], which probably make up a subset of the neurons we recorded. Future experiments utilizing combined Cre- and Flp-driver lines, such as those described in the present study, will help to elucidate the specificity of connectivity on inputs on to these cell types.

This general organization of a weak inhibition on to dopamine neurons and a strong inhibition on to local GABA neurons could potentially generate a pattern of feed-forward disinhibition, whereby dopamine neurons are briefly hyperpolarized by direct GABA input and then are disinhibited by a reduction in local GABA tone. The initial hyperpolarization may be a critical feature whereby dopamine neurons are poised to respond to disinhibition with a stronger burst of action potentials[42]. Indeed, this model is supported by our data showing strong induction of cFos in dopamine neurons after optogenetic activation of specific GABAergic inputs.

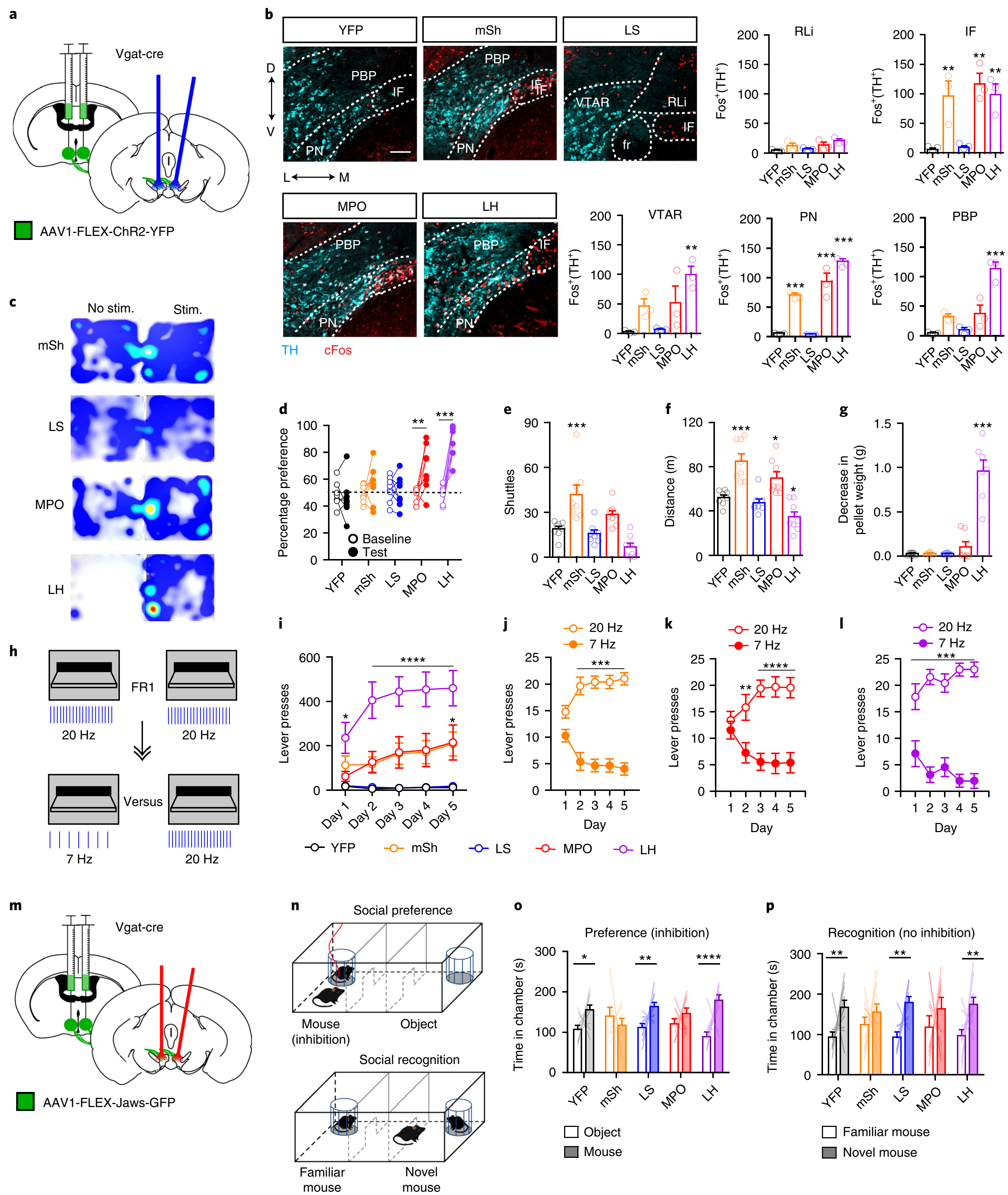
To determine connectivity in the present study, we measured only fast GABA transmission through $GABA_A$-receptors. Recent studies examining NAc-to-VTA inputs have also identified metabotropic $GABA_B$ currents, primarily on dopamine neurons[11,33]. It is possible that other GABAergic inputs to the VTA also activate $GABA_B$-receptors; however, for nearly every input previously examined, including the NAc, the net effect of stimulating the incoming GABAergic axons is disinhibition of dopamine neurons and increased dopamine release, resulting in reward reinforcement[15,30–34,43] and indicating that the $GABA_A$ current on local GABA neurons is the predominating outcome.

Distal GABAergic inputs from a variety of subcortical regions, many of which show clear innervation bias to different VTA subregions, are well set up to encode a wide variety of precise, behaviorally relevant information. Under this model, ascending cholinergic and glutamatergic projections may play a more permissive role in

Fig. 6 | Optogenetic activation and inhibition of specific distal GABAergic inputs differentially affect dopamine neuron activation and dopamine-dependent behaviors. a, Schematic of bilateral ChR2-YFP injection into a Vgat-Cre mouse and bilateral optic fiber implantation above the VTA. **b**, Example images showing staining for TH (cyan) and cFos (red) in the VTA after 20-Hz light stimulation (scale bar, 100 µm), and quantification of Fos$^+$ *Th*$^+$ cells in each VTA subregion in Vgat-Cre mice expressing YFP or ChR2-YFP in the indicated brain region ($n = 3$ mice per group; one-way ANOVA: VTAR: $F_{(4,10)} = 6.644$, $P = 0.007$; IF: $F_{(4,10)} = 10.72$, $P = 0.001$; PN: $F_{(4,10)} = 62.98$, $P < 0.0001$; PBP: $F_{(4,10)} = 25.87$, $P < 0.0001$; Dunnett's two-tailed selected comparisons test versus YFP: $^{**}P < 0.01$, $^{***}P < 0.001$). **c**, Example heat plots showing time spent in no stimulation (No stim.) versus stimulation (Stim.) side of the RTPP arena. **d**, Percentage of time spent in paired chamber during the 10-min baseline pretest or during the 20-min light-paired session ($n = 8$ mice per group; two-way RM ANOVA: $F_{(4,35)} = 11.56$, $P < 0.0001$; Bonferroni's multiple comparisons test: $^{**}P < 0.01$, $^{***}P < 0.001$). **e**, Number of shuttles between unpaired and paired chambers during light-paired session ($n = 8$ mice per group; one-way ANOVA: $F_{(4,35)} = 13.69$, $P < 0.0001$; Dunnett's two-tailed selected comparisons test versus YFP: $^{***}P < 0.001$). **f**, Distance traveled during light-paired session ($n = 8$ mice per group, one-way ANOVA: $F_{(4,35)} = 17.47$, $P < 0.0001$; Dunnett's two-tailed selected comparisons test versus YFP: $^{*}P < 0.05$, $^{***}P < 0.001$). **g**, Decrease in food pellet weight placed into empty cage with each mouse after 20 min of 20-Hz stimulation ($n = 8$ mice for YFP and mSh, 7 mice for LS, MPO and LH; one-way ANOVA: $F_{(4,32)} = 48.93$, $P < 0.0001$; Dunnett's two-tailed selected comparisons test versus YFP: $^{***}P < 0.001$). **h**, Schematic of lever pressing for light stimulation. Mice received 5 d of FR1 training in which either lever delivered 20-Hz stimulation, followed by 5 d of training in which their preferred lever was switched to 7-Hz stimulation. **i**, Number of lever presses per 1-h session during 20-Hz acquisition phase ($n = 8$ for YFP, mSh and LS; $n = 7$ for MPO and LH; two-way RM ANOVA: $F_{(16,132)} = 7.80$, $P < 0.0001$; Bonferroni's multiple comparisons test: $^{*}P < 0.05$, $^{****}P < 0.0001$ versus YFP). **j–l**, Lever presses on high- or low-frequency levers (25 trials) for mSh, MPO and LH mice ($n = 8$ for mSh, $n = 7$ for MPO and LH; two-way RM ANOVA: mSh $F_{(4,56)} = 17.81$ $P < 0.0001$; MPO $F_{(4,48)} = 18.09$, $P < 0.0001$; LH $F_{(4,48)} = 6.34$, $P = 0.0004$; Bonferroni's multiple comparisons test: $^{**}P < 0.01$, $^{***}P < 0.001$, $^{****}P < 0.0001$). **m**, Schematic of bilateral Jaws-GFP injection into a Vgat-Cre mouse and bilateral optic fiber implantation above the VTA. **n**, Schematic of social preference and social recognition assays. Mice received red light stimulation while in the mouse chamber during the preference assay. **o**, Time spent in the mouse or object chamber during the 5-min trial period ($n = 11$ mice for YFP and LH, $n = 10$ mice for mSh, LS and MPO; two-way ANOVA Interaction: $F_{(4,94)} = 4.76$, $P = 0.002$; mouse versus object: YFP $^{*}P < 0.05$; LS $^{**}P < 0.01$, LH $^{****}P < 0.0001$, Fisher's two-tailed single comparisons test). **p**, Time spent in the novel or familiar mouse chamber during the 5-min trial period ($n = 11$ mice for YFP and LH, $n = 10$ mice for mSh, LS and MPO; two-way ANOVA chamber: $F_{(1,94)} = 26.99$, $P < 0.0001$; novel mouse versus familiar mouse: YFP $^{**}P < 0.01$; LS $^{**}P < 0.01$, LH $^{**}P < 0.01$, Fisher's two-tailed single comparisons test). Error bars represent s.e.m.

helping to set the rate of tonic dopamine firing and enabling bursting when coupled with disinhibition of local interneurons.

We did not observe significant effects of distal GABA inactivation on Pavlovian conditioning, although it should be noted that reducing phasic activation of dopamine neurons by ~70%, through genetic inactivation of NMDA-type glutamate receptors, also does not affect Pavlovian reward association[44]. This suggests potentially important redundant mechanisms, whereby either glutamatergic drive on to dopamine neurons or disinhibition through distal GABAergic inputs on to local GABA neurons is sufficient to provide the minimal amount of dopamine required for this behavior.



Local, but not distal, GABA KO disrupted conditioned threat discrimination indicating that GABAergic tone on to dopamine neurons plays an important role in this process. Consistent with this observation, glutamatergic inputs from the LH and BNST, which predominantly synapse on to VTA GABA neurons, are both aversive[19,20], and BNST glutamatergic inputs have been shown to respond to both foot shock and shock-associated cues[19]. Thus, glutamatergic inputs on to both VTA GABA neurons and dopamine neurons are likely to work in coordination in regulating threat discrimination.

We observed that inactivation of distal GABAergic inputs to the VTA disrupted discrimination between high and low rewards, suggesting that both local and distal GABAergic inputs are critical for value-based decision-making. Several studies have investigated individual GABAergic inputs to the VTA[11,19–21], but whether these inputs have similar or distinct functions has not been thoroughly investigated. We observed that LH inputs, which were the most broadly distributed throughout the VTA subregions, were the most strongly reinforcing and promoted a distinct gnawing phenotype. This is consistent with previous reports of optical stimulation of LH to VTA GABA projections causing robust place preference, self-stimulation behavior, and either feeding or gnawing[20,37,45]. Our observation that LH inputs, which broadly innervate the VTA, are more reinforcing than either the MPO or NAc mSh inputs is also consistent with our recent observations that simultaneous activation of dopamine neurons in the lateral and medial parts of the VTA is required for optimal reinforcement[46].

We observed that inactivation of distal GABAergic inputs resulted in impaired social behavior that was not observed in local GABA KO mice. This suggests that distal GABA projections are important for disinhibiting local GABA to enable a dopamine signal for regulating social behaviors[47]. Indeed, we found that inhibition of specific inputs from the MPO or NAc mSh were able to disrupt social preference and social recognition, whereas LH and LS inputs did not affect this behavior, further supporting distinct roles for GABAergic inputs to the VTA.

The role of the rostral VTA in reward behaviors remains somewhat elusive. We did not observe any behavioral effects of activating LS inputs, which predominantly innervate the VTAR. We also did not observe a significant increase in cFos after activation of LS GABA inputs, although it is unclear whether this is a result of the relative strength of this input compared with others examined, or whether the connectivity of this input is distinct and less likely to lead to cFos activation via disinhibition. However, evidence suggests that LS inputs to the VTAR do lead to dopamine neuron disinhibition[48].

The precise behavioral role of glutamatergic inputs from the mPFC to the VTA remains unclear. We found that these inputs showed biased innervation to different VTA subregions, as well as unique synaptic properties including a persistent facilitation. These properties are strikingly different from those of PPTg inputs, which were depressed by repetitive stimulation. This variability of basic synaptic properties may facilitate different roles for different glutamatergic inputs in the control of dopamine neuron firing, particularly during learning-dependent tasks. Consistent with an important role for plasticity in mPFC inputs to the VTA for regulating associative behavior, inactivation of NMDA-type glutamate receptors in mPFC projections to the VTA potently reduces simple Pavlovian reward association[49]. Future studies designed to interrogate the interactions between mPFC inputs to the medial VTA PN region and GABAergic disinhibitory inputs to this region will provide important new insights into the coordinated regulation of specialized dopamine projection neurons.

Online content

Any methods, additional references, Nature Research reporting summaries, source data, extended data, supplementary information, acknowledgements, peer review information; details of author contributions and competing interests; and statements of data and code availability are available at https://doi.org/10.1038/s41593-020-0657-z.

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Methods

Mice. All procedures were approved and conducted in accordance to the guidelines of the Institutional Animal Care and Use Committee of the University of Washington. Mice were housed on a 12-h light:dark cycle with freely available food and water. Approximately equal numbers of male and female mice were used for all experiments. Mice were group housed except during day/night locomotor experiments (see Supplementary Table 1 for information on mouse lines).

Viruses. All CAV and AAV viruses were produced in house with titers of 1–3 × 10^{12} particles ml^{-1} as described[24].

Surgery. Mice were anesthetized with isoflurane before and during viral injection. Mice recovered for at least 2 weeks (AAV) or 3 weeks (CAV) before experimentation. For slice electrophysiology, mice were injected at age approximately 5–6 weeks. For all other experiments, mice were injected at age 8–12 weeks (see Supplementary Table 2 for injection coordinates). Values are in millimeters, relative to bregma. *Y* values caudal to bregma were adjusted for bregma–lambda distance using a correction factor of 4.21 mm. For *Z* values the syringe was lowered 0.5 mm past the indicated depth and raised up at the start of the injection.

Note that we found that targeting the midpoint of the LH (−1.25 mm from bregma) resulted in robust Syn-GFP labeling and synaptic connectivity in Vgat-Cre mice. However, in Vglut2-Cre mice we observed consistent Li-EPSCs in slices only when targeting the most rostral aspect of the structure. This is consistent with previous reports observing functional LH–VTA glutamatergic connectivity, which also targeted the rostral LH[20].

Retrograde input mapping. At least 3 weeks after injection of 500 nl of CAV2-FLEX-zsGreen virus into the VTA, mice were euthanized and perfused with 4% paraformaldehyde. Then 30-µm frozen brain sections were collected and mounted on glass slides. One brain section per atlas page[25] was imaged at ×10 magnification using a Keyence BZ-X710 fluorescent microscope, and cells in each brain region were counted by an experienced investigator. Brain regions were included if we observed retrograde labeling in at least half the injected animals.

Syn-GFP input mapping. At least 2 weeks after injection of AAV1-FLEX-Syn-GFP (300 nl), mice were euthanized and perfused with 4% paraformaldehyde. Then 30-µm frozen brain sections were collected. One section per atlas image for the rostral–caudal extent of the VTA was selected and stained overnight with a rabbit anti-GFP antibody (Invitrogen, catalog no. A11122, 1:2,000). Images were collected at ×10 magnification using a Keyence BZ-X710 fluorescent microscope and analyzed using ImageJ software. Images were background subtracted, and mean pixel intensity and integrated pixel density were measured for each VTA subregion. For normalized intensity plots, all subregions from an individual animal were normalized to the highest intensity subregion from that animal.

RNAscope in situ. For RNAscope in situ combined with RetroBeads, mice were stereotaxically injected in the VTA with red RetroBeads (Lumafluor); 2 weeks after injection the brains were collected and snap frozen, and 20-µm sections were mounted on to slides. The RNAscope v.1 assay (Advanced Cell Diagnostics) was performed according to manufacturer's directions using probes directed against *Slc32a1* (Vgat), *Slc17a6* (Vglut2) or *Slc17a8* (Vglut3). For in situ assessment of gene KO, *Vgat$^{loxP/loxP}$* mice were injected into the VTA with AAV1-ΔCre-GFP, AAV1-Cre-GFP or CAV-Cre+AAV1-FLEX-Vgat. Then 3 weeks after injection the brains were collected and snap frozen, and 20-µm sections were mounted on to slides. The RNAscope v.2 assay (Advanced Cell Diagnostics) was performed according to manufacturer's directions using probes directed against *Cre* and *Slc32a1* (Vgat).

Slice electrophysiology. Horizontal or coronal brain slices (200 or 250 µm, respectively) were prepared in an ice slush solution containing (in mM): 92 NMDG, 2.5 KCl, 1.25 NaH_2PO_4, 30 $NaHCO_3$, 20 4-(2-hydroxyethyl)-1-piperazine-ethanesulfonic acid (Hepes), 25 glucose, 2 thiourea, 5 sodium ascorbate, 3 sodium pyruvate, 0.5 $CaCl_2$ and 10 $MgSO_4$, pH 7.3–7.4 (ref. [50]). Slices recovered for ≤12 min in the same solution at 32 °C and were then transferred to a room temperature solution including (in mM): 92 NaCl, 2.5 KCl, 1.25 NaH_2PO_4, 30 $NaHCO_3$, 20 Hepes, 25 glucose, 2 thiourea, 5 sodium ascorbate, 3 sodium pyruvate, 2 $CaCl_2$ and 2 $MgSO_4$. Slices recovered for an additional 45 min before recordings were made in artificial cerebrospinal fluid at 32 °C, continually perfused over slices at a rate of ~2 ml min^{-1} and containing (in mM): 126 NaCl, 2.5 KCl, 1.2 NaH_2PO_4, 1.2 $MgCl_2$ 11 D-glucose, 18 $NaHCO_3$ and 2.4 $CaCl_2$. All solutions were continually bubbled with O_2/CO_2.

Whole-cell recordings were made using an Axopatch 700B amplifier (Molecular Devices) with filtering at 1 kHz using 4- to 6-MΩ electrodes. For mIPSC recordings electrodes were filled with an internal solution containing (in mM): 135 KCl, 12 NaCl, 0.5 (ethylenebis(oxonitrilo))tetra-acetate (EGTA), 10 Hepes, 2.5 Mg^{2+}/ATP, 0.25 Na^+-GTP, pH 7.3, 280 mosmol. For Li-EPSC, Li-IPSC and action potential recordings, electrodes were filled with an internal solution containing (in mM): 130 potassium gluconate, 10 Hepes, 5 NaCl, 1 EGTA), 5 Mg^{2+}/ATP, 0.5 Na^+-GTP, pH 7.3, 280 mosmol.

The mIPSC recordings were made with holding at −60 mV in the presence of tetrodotoxin (500 nM) and kynurenic acid (2 mM). Events were analyzed using MiniAnalysis software (Synaptosoft) and automated detection with visual confirmation by an experienced investigator.

Light-evoked synaptic transmission was induced with 5-ms light pulses delivered at 0.1 Hz from an optic fiber placed directly in the bath. Li-EPSCs were measured with holding at −60 mV in the presence of picrotoxin (100 µM). Li-IPSCs were measured with holding at −30 mV. Amplitudes were calculated from an average of at least 10 events. Spontaneous action potentials were recorded in current clamp mode in the same neurons used for EPSC/IPSC recordings. Then 20-Hz trains of 5-ms light pulses were delivered for 1 s (20 pulses) once every 10 s to determine the effect of light on action potential firing rates. Pre- and post-train firing rates were calculated for the 2 s immediately before and after the light train, respectively, and were an average of at least three sweeps.

The mIPSC recordings and Li-EPSC recordings for Vglut1 and Vglut2 were performed in mice expressing FlpO recombinase under the control of the TH promoter. Dopamine and nondopamine neurons were identified by the presence or absence (respectively) of a Flp-dependent fluorescent marker delivered by AAV (AAV1-FLEXfrt-mCherry). Li-IPSC recordings were performed in Vgat-Cre mice, and GABA- and non-GABA neurons were identified by expression of a Cre-dependent fluorescent marker delivered by AAV (AAV1-FLEX-GFP).

Behavioral assays in *Vgat$^{loxP/loxP}$* mice. *Locomotion.* Mice were singly housed in standard cages with food and water provided that were placed inside infrared locomotion chambers (Columbus Instruments). Ambulatory activity (beam breaks) was measured in 15-min bins continuously for 3 nights and 2 d.

Cocaine sensitization. Mice were placed into a standard cage inside infrared locomotion chambers (Columbus Instruments). The cage was clean on the first day, and the same cage was used for all subsequent days. After 90 min, mice were injected subcutaneously with saline or 20 mg kg^{-1} of cocaine. Mice received 2 d of saline injections followed by 5 d of cocaine injections. The second saline day is plotted in Fig. 5b. Ambulatory activity (beam breaks) was measured in 5-min bins.

Pavlovian conditioning. Mice were food restricted to 85% of their free body weight. Mice were placed into operant conditioning boxes (Med Associates) and received 25 trials per day for 7 d. Each trial consisted of both levers extending (an auditory and visual cue) and remaining extended for 10 s, at which point both levers retracted and a single 20-mg reward pellet was delivered (unflavored Purified Dustless Precision Pellets, Bio-Serv). Trials were separated by a variable intertrial interval (ITI) averaging 60 s. Head entries during the cue period and the ITI were measured using an infrared detector in the food hopper.

Instrumental responses. After Pavlovian training, mice were placed in the same operant boxes. Both levers extended and remained extended until a press was made on either lever, at which point both levers would retract and one pellet was delivered. Levers would re-extend when the mouse made a head entry into the food hopper. Mice were given a 1-h session each day for 3 d.

Lever switching. Each mouse's nonpreferred lever was determined based on their performance on day 3 of instrumental training. The nonpreferred lever was set as the high-reward lever (three pellets) and the preferred lever was set as the low-reward lever (one pellet). Mice were given 25 trials in which both levers extended; after a press on either lever, both levers would retract and the appropriate reward was delivered, followed by a 1-min ITI before the levers extended for the next trial.

Fear conditioning. Fear conditioning was performed as described[35]. Briefly, mice were placed into context A and baseline freezing was measured across three presentations of the CS^+ tone and three presentations of the CS^- tone. Tones were counterbalanced across groups. Mice then received 2 d of conditioning in context B, consisting of 10 presentations of the CS^+ tone (10 s) co-terminating with a 0.5-s foot shock (0.3 mA) and 10 presentations of the CS^- tone. The day after the second conditioning session, mice were returned to context A and freezing was measured during three presentations of each CS tone. Movement was tracked using Ethovision software (Noldus) and freezing calculated using a custom Matlab script.

Three-chamber social assay. The arena was a white Plexiglas box (60 × 30 × 30 cm^3) divided into three equal-sized chambers with clear Plexiglas dividers, each with a doorway allowing the mice to pass freely between chambers. Mice were given 10 min to explore the empty arena, then were briefly removed and returned to their home cage while the novel object (empty wire pencil cup) was introduced to one chamber and the first mouse (contained in a wire pencil cup) was introduced to the opposite chamber. The experimental animal was returned to the arena for a 30-min exploration, of which the first 5 min were scored, and then briefly removed again while the novel mouse was added, before a final 5-min exploration.

Optogenetic behavioral experiments. Vgat-Cre mice were injected bilaterally in the indicated region with 400 nl of AAV1-FLEX-ChR2-YFP or AAV1-FLEX-Jaws-GFP. Control mice were injected with AAV1-FLEX-YFP. Optic fibers were implanted bilaterally above the VTA, with one fiber implanted at a 10° angle to allow enough working room between the ferrules to attach patch cables. Histology to verify targeting was performed on ChR2 and Jaws mice only.

Real-time place preference/aversion: on day 1 mice were placed into a two-chambered arena and allowed to explore freely for 10 min. The walls of the arena had horizontal black and white stripes in one chamber and vertical black and white stripes in the other chamber. Mice were then assigned a light-paired chamber such that any inherent side bias was canceled out within groups. On day 2 mice were connected to patch cables and placed into the unpaired chamber to begin the trial. A 20-Hz, 5-ms blue light stimulation (ChR2) or red light stimulation of 2 s on, 1 s ramp down and 1 s off (Jaws) was delivered whenever the centerpoint of the mouse was in the paired chamber. The trial lasted for 20 min.

Intracranial self-stimulation. Mice were food restricted to 85% of free body weight to increase exploratory activity. Mice were given a 1-h session each day in which both levers extended and a press on either lever led to lever retraction and 3 s of 20-Hz light stimulation. Levers re-extended after an additional 2-s timeout period. After 5 d of lever acquisition the frequency of light stimulation was decreased to 7 Hz on each mouse's preferred lever, whereas the frequency on the nonpreferred lever remained at 20 Hz.

Light-induced gnawing behavior. Mice were placed in a clean empty cage with no bedding and given a single standard food pellet, which was weighed before the trial. Mice were then given 20 min of 20-Hz light stimulation and the food pellet was weighed again at the end of the trial.

Three-chamber social assay. Mice were connected to the patch cables and placed into the empty three-chamber arena for a 10-min habituation period. Next, the experimental mice were briefly removed from the chamber, but remained connected to the patch cables while the object (empty wire pencil cup) and first mouse (under a pencil cup) were introduced to opposite chambers. Experimental mice were returned to the arena for a 30-min period during which they received red light stimulation (2 s on, 1 s ramp down, 2 s off) whenever they were in the chamber containing the mouse. The first 5 min of this period were scored for social preference. The experimental mouse was then briefly removed from the chamber while the second mouse was added under the empty pencil cup, and the experimental mouse was returned for a final 5-min period. The mouse remained connected to the patch cables but no light stimulation was given during this period.

Staining for cFos. Mice were given 20 min of 20-Hz light stimulation and were perfused 1 h after the end of the stimulation period. Then 30-μm sections were collected and one section per atlas image for the rostral–caudal extent of the VTA was selected and stained overnight at room temperature, with antibodies against cFos (Millipore, catalog no. ABE457, 1:1,000) and TH (Millipore, catalog no. MAB318, 1:1,000). Images were collected at ×20 magnification using a Keyence BZ-X710 fluorescent microscope and analyzed using ImageJ software.

Statistics and reproducibility. All data were analyzed for statistical significance using GraphPad Prism software. For multiple groups or multiple measures, we used one-way or two-way analysis of variance (ANOVA). Tukey's multiple comparison tests were used for one-way ANOVAs with multiple comparisons across groups. Dunnett's selected comparisons were used for directly comparing experimental groups with the control group. Bonferroni's multiple comparison tests were used for between-subject comparisons after two-way ANOVA and Fisher's single comparison tests were used for a within-subject comparison. Data distribution was assumed to be normal but this was not formally tested. No statistical methods were used to predetermine sample sizes, but our sample sizes are similar to those reported in previous publications[35,46]. All behavioral assays were repeated in a minimum of three cohorts with similar replication of results. Data for imaging and electrophysiology experiments were pooled from at least three individual animals and collected from at least two rounds of experiments, with similar replication of results. Littermates were randomly assigned to experimental groups and animals were tested in a random order. Data were analyzed by an investigator blinded to experimental condition; the experimenter was not blinded during data collection. Animals with missed viral injections or significant viral spread outside the targeted region were excluded from analyses.

Reporting summary. Further information on research design is available in the Nature Research Reporting Summary linked to this article.

Data availability
Datasets supporting the findings in the present study are available from the corresponding author upon reasonable request. All viral vectors used in this manuscript are available from the corresponding author upon reasonable request.

Code availability
Code used for behavioral analysis is available from the corresponding author upon reasonable request.

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Author contributions
M.E.S., A.S.C., J.M.R. and L.S.Z. designed the experiments, and collected and analyzed the data. M.E.S., A.S.C. and J.M.R. performed the viral injection surgery. M.E.S. and B.C. performed the behavioral analysis. M.E.S. performed the slice electrophysiology. M.E.S., A.S.C., B.C., J.M.R. and L.S.Z. performed the histology and cell counts. L.S.Z. generated CAV2-FLEX-ZsGreen. L.S.Z. and M.E.S purified all viral vectors. R.A. provided the Th^{FlpO} mouse line. M.E.S. and L.S.Z. wrote the manuscript.

Competing interests
The authors declare no competing interests.

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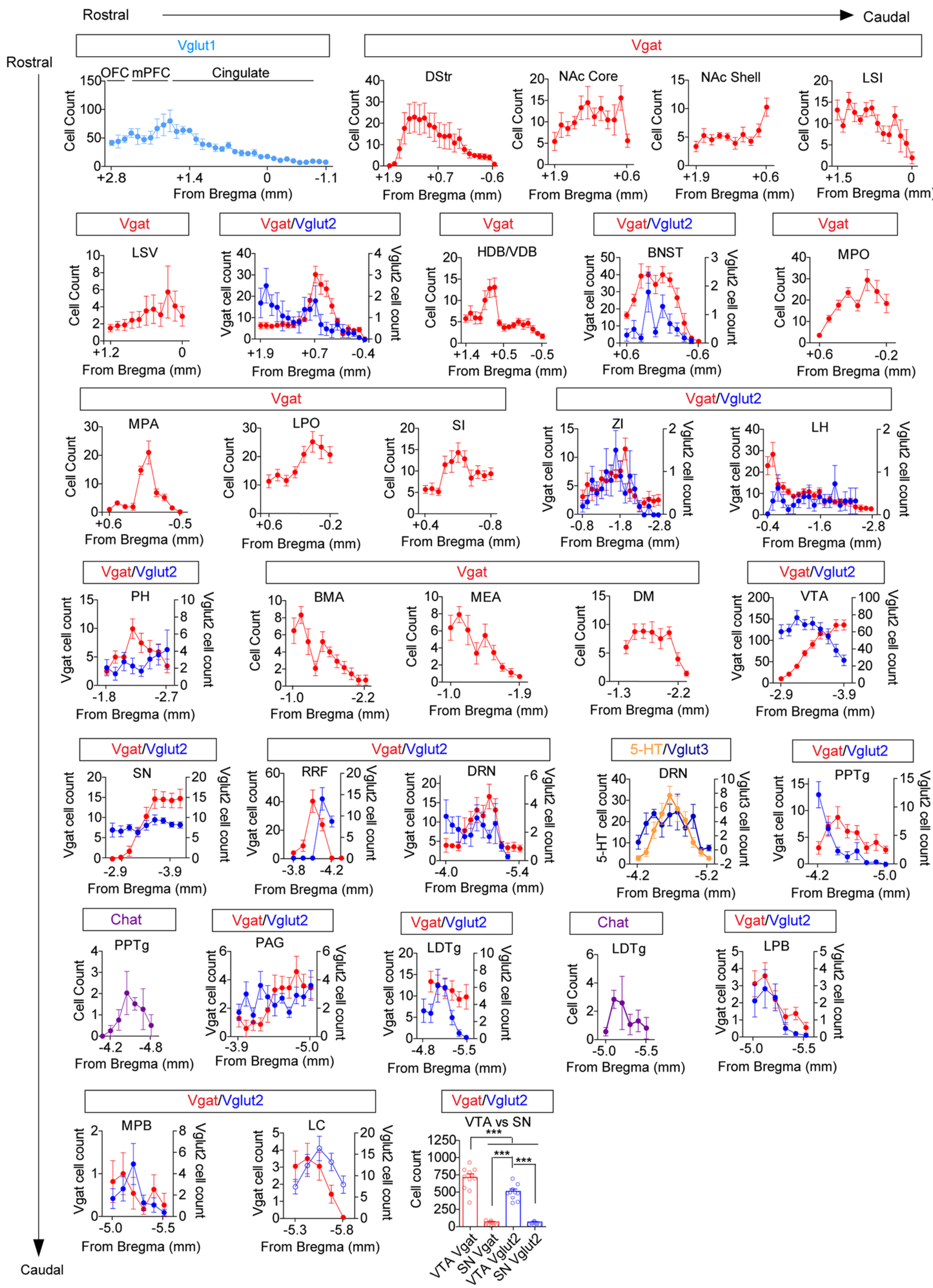
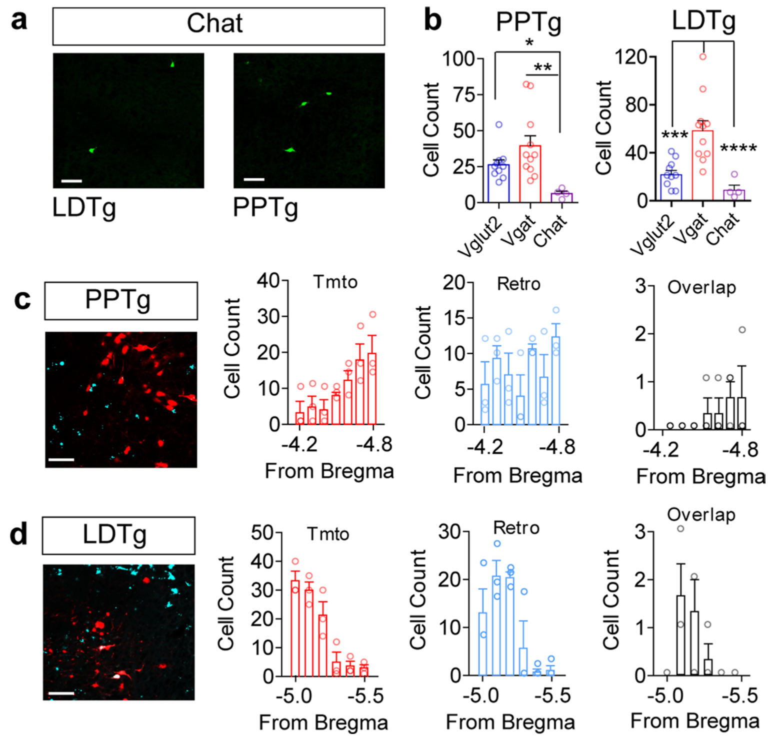
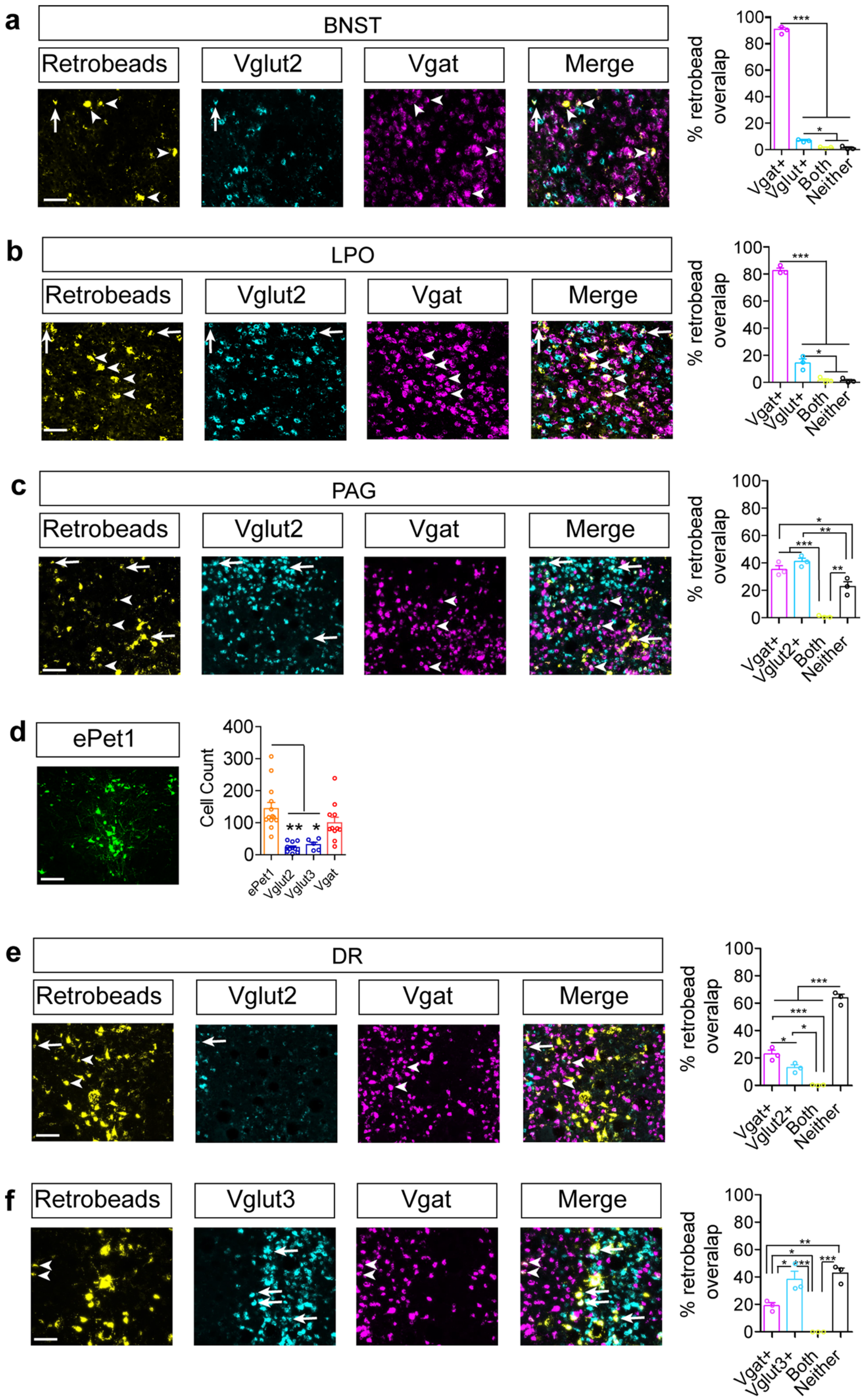


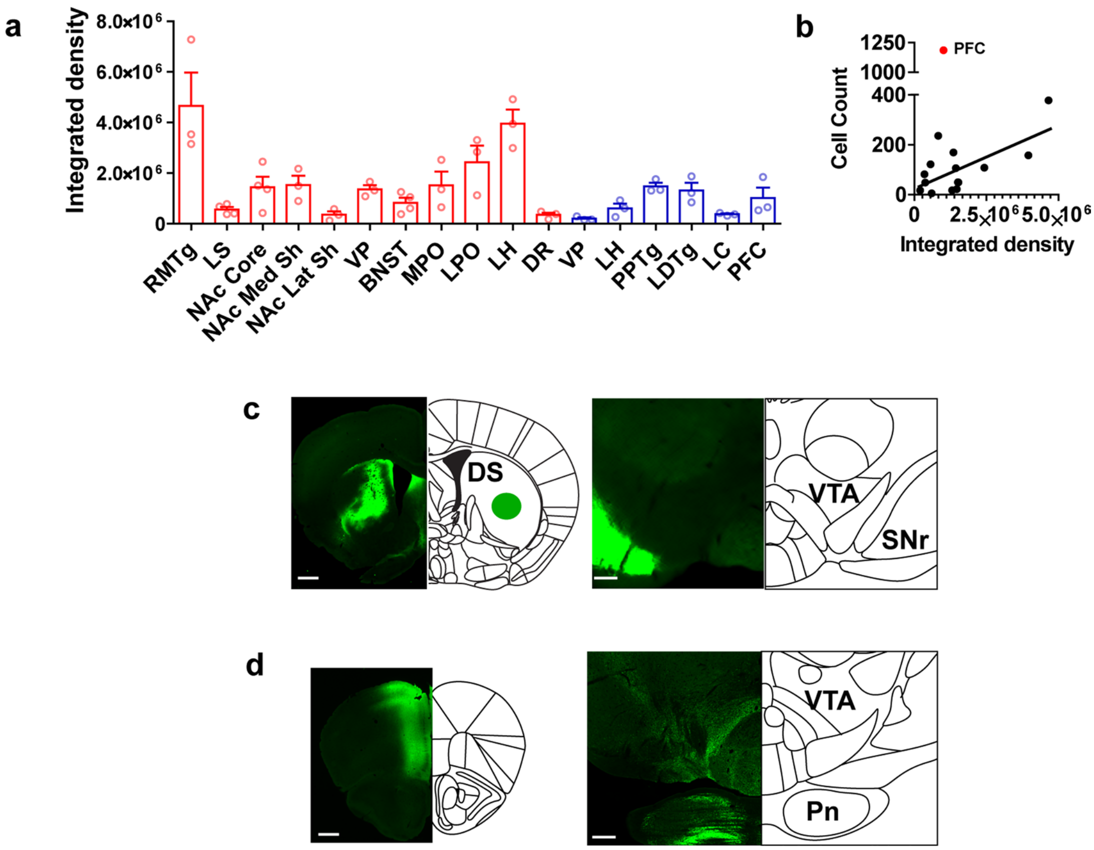
Extended Data Fig. 1 | Distribution of CAV2-FLEX-zsGreen retrogradely labeled neurons. Cell counts of retrogradely labeled cells in indicated Cre lines across the rostral-caudal axis. Note that cells from different Cre lines in the same region are plotted on different axes. (Vgat n = 11 mice, Vglut1 n = 4; Vglut2 n = 10; Vglut3 n = 5, 5-HT n = 13, Chat n = 4.) Final panel depicts counts of infected cells in the VTA and neighboring substantia nigra (SN) region (Vgat n = 11 mice, Vglut2 n = 10 mice, One way ANOVA $F_{(3,38)}$=112.3 p < 0.0001, Tukey's Multiple Comparison ***p < 0.001). Error bars represent s.e.m.



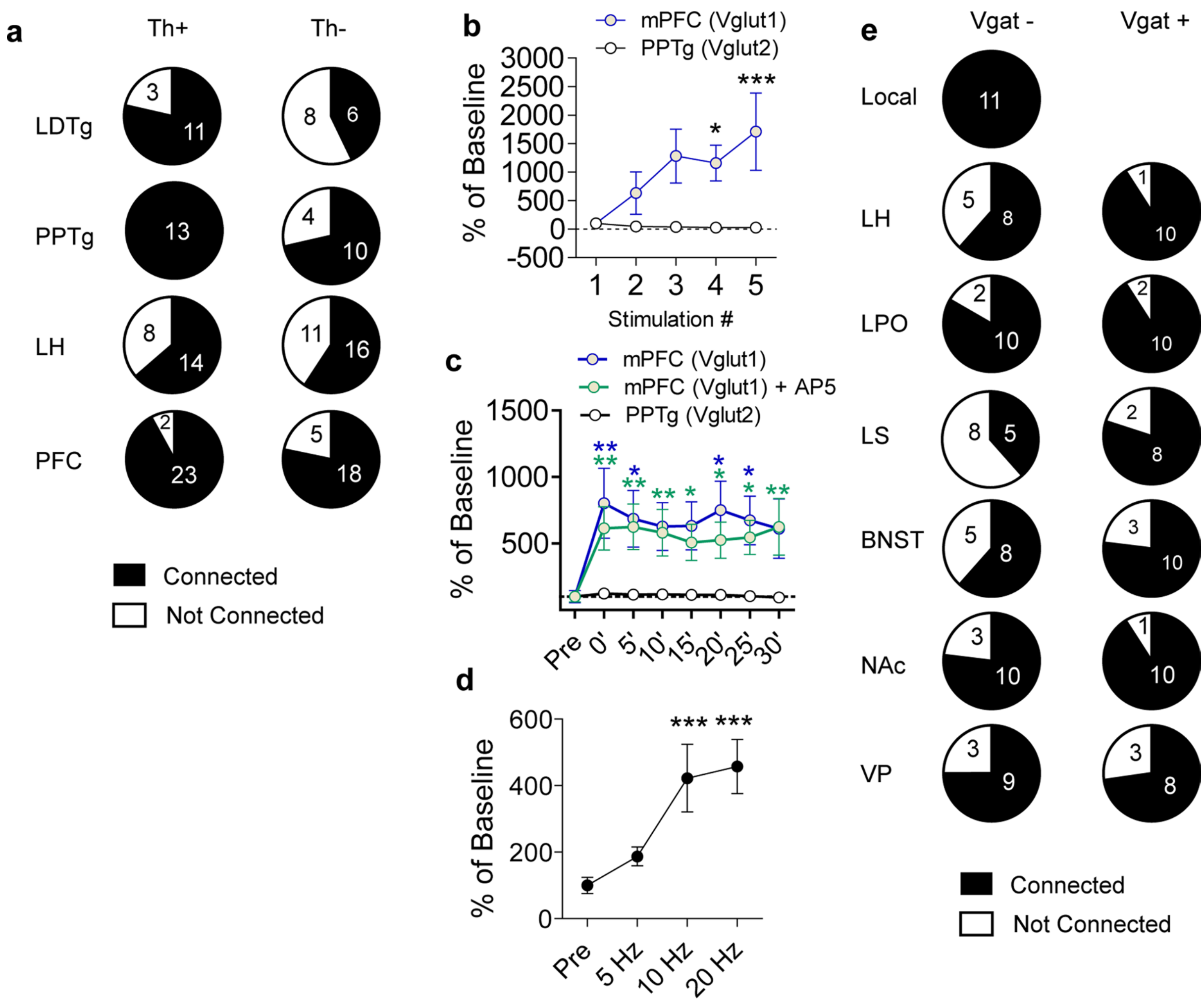
Extended Data Fig. 2 | Retrobead labeling of cholinergic neurons in the PPTg and LDTg. (a) Representative image of CAV2-FLEX-zsGreen retrogradely labeled cells in the PPTg and LDTg of a Chat Cre mouse. **(b)** Total cells labeled in the PPTg and LDTg in different Cre driver lines (Vglut2 n = 10, Vgat n = 11, Chat n = 4; PPTg, One-way ANOVA, $F_{(2,28)}$=7.740, P = 0.0023, *P < 0.05, **P < 0.01; LDTg, One-way ANOVA, $F_{(2,28)}$=17.41, P < 0.0001, ***P < 0.01, ****P < 0.0001). **(c, d)** Representative images of Chat Cre:Tomato cells (red) and RetroBeads from the VTA (cyan) in the PPTg and LDTg, and quantification of cell counts and overlap across the rostral-caudal axis (n = 3/group). Error bars represent s.e.m. Scale bars = 50µm.



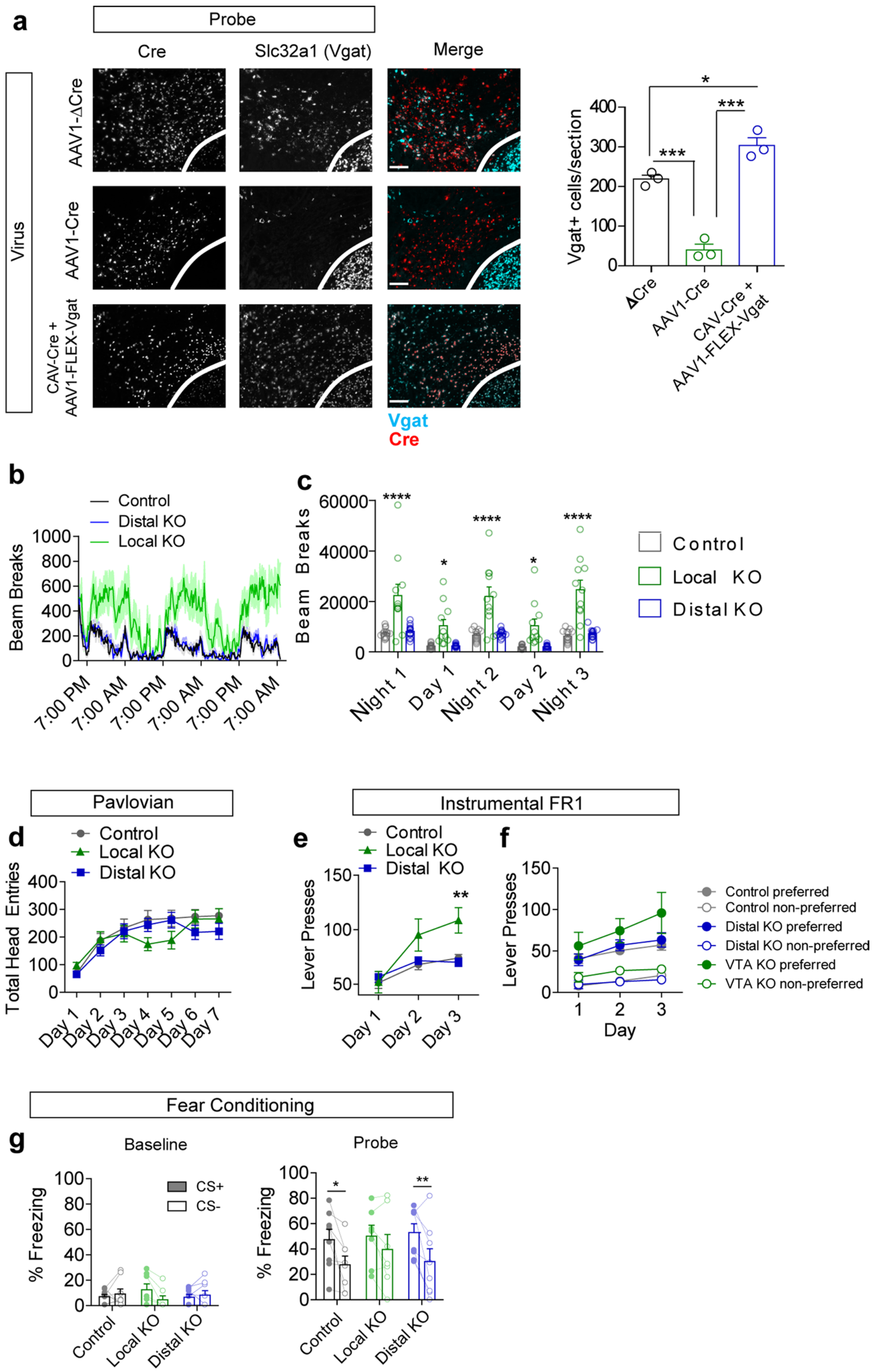
Extended Data Fig. 3 | Retrobead and *in situ* labeling of VTA inputs. (a–c) Representative images of indicated regions containing RetroBeads transported from the VTA and RNAscope *in situ* against *Slc32a1* (Vgat) or *Slc17a6* (Vglut2) and quantification of the percent of RetroBead labeled cells that colabel with *in situ* probes for BNST(a), LPO(b), and PAG(c). Arrows identify RetroBead-labeled Vglut-positive neurons, while arrowheads identify RetroBead-labeled Vgat-positive neurons (n = 3 mice, cells counted in 3-5 sections per region per mouse. One-way ANOVA, BNST: $F_{(3,8)}$=2140, P < 0.0001, LPO: $F_{(3,8)}$=441.3, P < 0.0001, PAG: $F_{(3,8)}$=48.31, P < 0.0001, Tukey's Multiple Comparisons *P < 0.05, **P < 0.01, ***P < 0.001). **(d)** Representative CAV2-FLEX-ZsGreen labeling in DR of ePet1-Cre mice (scale bar = 100 µm) and cell counts in DR for ePet1, Vglut2, Vglut3, and Vgat Cre mice (ePet n = 13, Vglut2 n = 10, Vglut3 n = 5, Vgat n = 11 mice; One-way ANOVA, $F_{(3,35)}$=12.10, P = 0.0001, Tukey's Multiple Comparisons *P < 0.05, **P < 0.01). **(e, f)** Representative images for DR RetroBeads transported from the VTA and RNAscope *in situ* against *Slc32a1* (Vgat), *Slc17a6* (Vglut2), or *Slc17a8* (Vglut3), and quantification of the percent of RetroBead labeled cells that colabel with *in situ* probes. Arrows identify RetroBead-labeled Vglut-positive neurons, while arrowheads identify RetroBead-labeled Vgat-positive neurons (n = 3 mice, cells counted in 3-5 sections per mouse. One-way ANOVA, Vglut2: $F_{(3,8)}$=161.9, P < 0.0001, Vglut3: $F_{(3,8)}$=28.94, P = 0.0001, Tukey's Multiple Comparisons *P < 0.05, **P < 0.01, ***P < 0.001). Error bars represent s.e.m. Scale bars = 50µm.



Extended Data Fig. 4 | Density of inputs to the VTA. (a) Integrated density (arbitrary units) of GFP fluoresence in the VTA following injection of synaptophysinGFP into the indicated region. Red bars: Vgat Cre, blue bars: Vglut 1 or 2 Cre (n = 3 mice/group for all regions except n = 4 mice for LS, NAc core, and BNST). **(b)** Correlation between average total integrated fluoresence density in the VTA and the average number of cells retrogradely labeled by CAV-FLEX-ZsGreen in each region. Black line = linear regression of all points excluding PFC (n = 3 mice/group for all regions except n = 4 mice for LS, NAc core, and BNST; Pearson r = 0.6570, Spearman two-tailed P = 0.0078). **(c)** Representative images of synaptophysinGFP injection into the dorsal striatum of a Vgat Cre mouse, and terminals in the SNr, adjacent to the VTA. Scale bars = 500 µm (left) and 200 µm (right). **(d)** Representative images of synaptophysinGFP injection into the PFC and terminals in the VTA-paranigral region and the adjacent pontine nucleus. Error bars represent s.e.m.

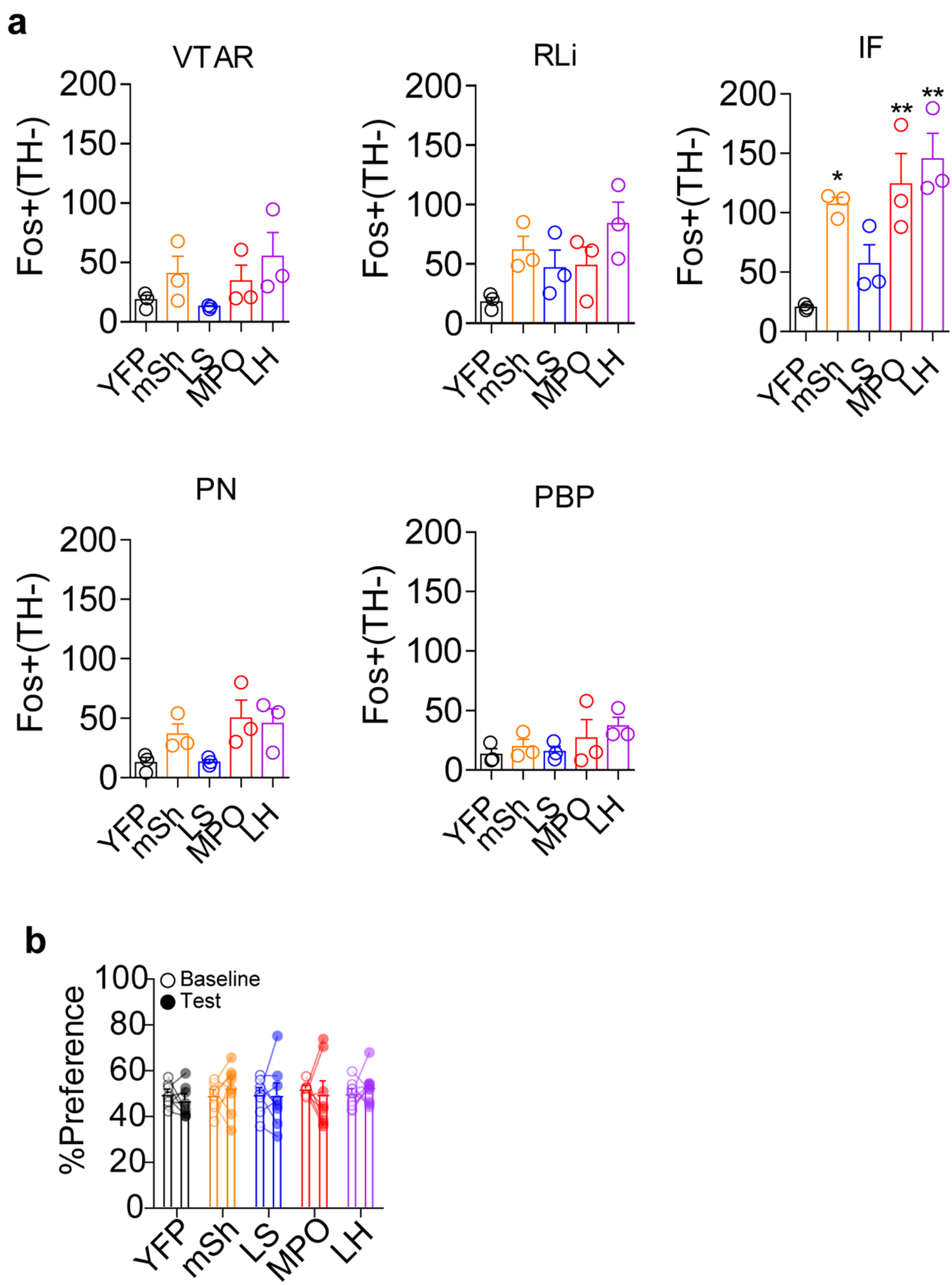


Extended Data Fig. 5 | Connectivity of VTA inputs. **(a)** Numbers of connected and not connected Th+ and Th- cells patched in the VTA with ChR2 expressed in the indicated region. Connected cells were those with a visible EPSC detectable across an average of 10 traces. Connected cells in the PFC include those cells that had no visible Li-EPSC until after high frequency stimulation. **(b)** Percent change in Li-EPSCs amplitude during the first five pulses of a 20 Hz train of light pulses activating PFC or PPTg inputs relative to the first pulse (Two-way RM ANOVA, $F_{(4,60)}$=3.280, P = 0.0178, Bonferroni multiple comparisons *P < 0.05, ***P < 0.001; PFC n = 8 cells, PPTg n = 9 cells). **(c)** Percent change in amplitude of Li-EPSCs before and after high frequency stimulation of PFC or PPTg inputs relative to pre-stimulus train amplitude. PFC inputs were stimulated in the presence or absence of AP5 (100 μM), which remained in the bath for the duration of the experiment. (Two-way RM ANOVA, $F_{(14,154)}$=2.40, P = 0.0046, Bonferroni multiple comparisons vs PPTg *P < 0.05, **P < 0.01; PFC n = 8 cells, PFC + AP5 = 7 cells, PPTg n = 9 cells). **(d)** Percent of baseline Li-EPSC amplitude following 3×1 s stimulus trains at the indicated frequencies (One-way RM ANOVA, $F_{(3,8)}$=13.13, P < 0.0001, Tukey's Multiple Comparisons ***P < 0.001 vs Pre, n = 9 cells). **(e)** Numbers of connected and not connected Vgat- and Vgat+ cells patched in the VTA with ChR2 expressed in the indicated region. Connected cells were those with a visible IPSC detectable across an average of 10 traces. Error bars represent s.e.m.



Extended Data Fig. 6 | See next page for caption.

Extended Data Fig. 6 | Local and distal Vgat knockout. (a) Example images of RNAscope *in situ* labeling *Cre* and *Slc32a1* (Vgat) in the VTA following injection of indicated viruses (red=Cre, cyan=Vgat), and quantification of Vgat+ cells/section (n = 3 mice/group, One way ANOVA $F_{(2,6)}$=77.97 p < 0.0001, Tukey's multiple comparisons *p < 0.05, ***p < 0.001, scale bar = 100 µm). **(b)** Locomotor activity (measured as infrared beam breaks) measured in 15 min bins (line) with s.e.m. (shading). **(c)** Locomotor activity summed over three nights (7 pm to 7 am) and two days (7 am to 7 pm) (Control n = 13 mice, VTA KO n = 11, Distal KO n = 15; Two-way RM ANOVA $F_{(8,144)}$=6.33, p < 0.0001; Bonferroni multiple comparisons *p < 0.05, ****p < 0.0001). **(d)** Total head entries during each day of Pavlovian training (n = 23 mice/group control, 21 VTA KO, 27 distal KO; 2-way RM ANOVA $F_{(12,402)}$=1.9, p = 0.032; Bonferroni multiple comparisons did not achieve significance). **(e)** Total lever presses during each day of FR1 instrumental conditioning (1 h session/day) (Control n = 21 mice, VTA KO n = 14 mice, Distal KO n = 27 mice; Two-way RM ANOVA $F_{(4,118)}$=5.35, p = 0.0005; Bonferroni multiple comparisons **p < 0.01). **(f)** Lever presses on the preferred and non-preferred levers during 3 days of FR1 instrumental conditioning (Control n = 15 mice, VTA KO n = 11 mice, Distal KO n = 15 mice). **(g)** Percent of time spent freezing during delivery of the CS + or CS- tone during a baseline pretest or following two days of fear conditioning (n = 8 for Control and Distal KO, n = 7 for Local KO; Probe: 2-way RM ANOVA significant effect of CS $F_{(1,20)}$=19.79 p = 0.0002, Bonferroni multiple comparisons *p < 0.05, **p < 0.01). Error bars represent s.e.m.



Extended Data Fig. 7 | Fos induction in TH- cells and Jaws RTPA. (a) Number of Fos+ TH- cells in each VTA subregion in Vgat-Cre mice expressing YFP or ChR2-YFP in the indicated brain region. (n = 3 mice/group, For IF: One-way ANOVA $F_{(4,10)}$=9.207, p = 0.002; *p < 0.05, **p < 0.01 vs. YFP.) **(b)** Real-time place aversion assay comparing percent of time spent in the light paired chamber during the pretest (baseline) period and during the light stimulation period for Vgat-Cre mice expressing YFP or Jaws-GFP in the indicated regions (n = 8 for all groups except n = 10 for LH). Error bars represent s.e.m.

nature research

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For all statistical analyses, confirm that the following items are present in the figure legend, table legend, main text, or Methods section.

n/a	Confirmed	
☐	☒	The exact sample size (*n*) for each experimental group/condition, given as a discrete number and unit of measurement
☐	☒	A statement on whether measurements were taken from distinct samples or whether the same sample was measured repeatedly
☐	☒	The statistical test(s) used AND whether they are one- or two-sided *Only common tests should be described solely by name; describe more complex techniques in the Methods section.*
☐	☒	A description of all covariates tested
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☐	☒	For null hypothesis testing, the test statistic (e.g. F, t, r) with confidence intervals, effect sizes, degrees of freedom and P value noted *Give P values as exact values whenever suitable.*
☒	☐	For Bayesian analysis, information on the choice of priors and Markov chain Monte Carlo settings
☒	☐	For hierarchical and complex designs, identification of the appropriate level for tests and full reporting of outcomes
☒	☐	Estimates of effect sizes (e.g. Cohen's *d*, Pearson's *r*), indicating how they were calculated

Our web collection on statistics for biologists contains articles on many of the points above.

Software and code

Policy information about availability of computer code

Data collection	Images were obtained using Keyence BZ-X viewer software (version 1.4.0.1). Electrophysiology data were collected using Clampex software (Version 10.3). Pavlovian and instrumental behavioral data were collected using MedAssociates software (version 4.2).
Data analysis	Image analysis was conducted using ImageJ (version 1.44). Slice electrophysiology data were analyzed using Clampfit (version 10.3.0.2) and MiniAnalysis (version 6.0.7). Behavioral data were analyzed using Ethovision XT (version 8.5). Freezing during fear conditioning was determined using a custom Matlab script (version R2019b). Statistical analyses were performed using Graphpad Prism (version 5).

For manuscripts utilizing custom algorithms or software that are central to the research but not yet described in published literature, software must be made available to editors/reviewers. We strongly encourage code deposition in a community repository (e.g. GitHub). See the Nature Research guidelines for submitting code & software for further information.

Data

Policy information about availability of data

All manuscripts must include a data availability statement. This statement should provide the following information, where applicable:

- Accession codes, unique identifiers, or web links for publicly available datasets
- A list of figures that have associated raw data
- A description of any restrictions on data availability

Datasets supporting the findings in this study are available from the corresponding author upon reasonable request.

Field-specific reporting

Please select the one below that is the best fit for your research. If you are not sure, read the appropriate sections before making your selection.

- [x] Life sciences
- [ ] Behavioural & social sciences
- [ ] Ecological, evolutionary & environmental sciences

For a reference copy of the document with all sections, see nature.com/documents/nr-reporting-summary-flat.pdf

Life sciences study design

All studies must disclose on these points even when the disclosure is negative.

Sample size	Power analysis was not used, instead sample sizes were based upon previously published variability of assays, including our own published experiments of a similar nature, and sufficient numbers for duplication.
Data exclusions	Animals were excluded from behavioral assays based on incorrect targeting of virus.
Replication	All behavioral assays were repeated in a minimum of three cohorts with successful replication of results. Data for imaging and electrophysiology experiments were pooled from at least three individual animals, collected from at least two rounds of experiments, with successful replication of results.
Randomization	All animals were arbitrarily assigned to control or experimental groups. The order of running animals was randomized across groups.
Blinding	Electrophysiology and behavioral data were analyzed by an investigator blind to the testing condition. The experimenter was not consistently blinded during data collection as the surgeon conducted many of the experiments.

Reporting for specific materials, systems and methods

We require information from authors about some types of materials, experimental systems and methods used in many studies. Here, indicate whether each material, system or method listed is relevant to your study. If you are not sure if a list item applies to your research, read the appropriate section before selecting a response.

Materials & experimental systems

n/a	Involved in the study
☐	☒ Antibodies
☒	☐ Eukaryotic cell lines
☒	☐ Palaeontology
☐	☒ Animals and other organisms
☒	☐ Human research participants
☒	☐ Clinical data

Methods

n/a	Involved in the study
☒	☐ ChIP-seq
☒	☐ Flow cytometry
☒	☐ MRI-based neuroimaging

Antibodies

Antibodies used	Rabbit anti GFP, Invitrogen A11122, 1:2000; Rabbit anti cFos, Millipore ABE457, 1:1000 Mouse anti TH, Millipore MAB318 Clone LNC1, 1:1000. Lot numbers are not available. Secondary antibodies (raised in donkey) were conjugated to DyLight488, CY3, or CY5 (1:250, Jackson Immunolabs).
Validation	Per the manufacturers' websites: Rabbit anti GFP (Invitrogen A11122) has been validated for use in IHC and referenced in at least 376 publications for this application. Rabbit anti cFos (Millipore ABE457) was validated for IHC in rat tissue (100% sequence homology to mouse). A literature search finds over 150 publications for this application. Mouse anti TH (Millipore MAB318) has been validated on PFA fixed frozen tissue from mouse and referenced in over 200 publications.

Animals and other organisms

Policy information about studies involving animals; ARRIVE guidelines recommended for reporting animal research

Laboratory animals	Male and female mice were used for this study. Mice were injected at 5-6 weeks of age for electrophysiology experiments. Mice were injected at 8-12 weeks of age for all other experiments. Strains of mice used were: Slc32a1-IRES-Cre, Slc17a7-IRES-Cre, Slc17a6-IRES-Cre, Slc17a8-iCre, Chat-IRES-Cre, Fev-Cre, Slc32a1 lox/lox, and Th-2A-FlpO.
Wild animals	The study did not involve wild animals.
Field-collected samples	The study did not involve samples collected from the field.

Ethics oversight	All animal procedures were approved and conducted in accordance to the guidelines of the Institutional Animal Care and Use Committee of the University of Washington.

Note that full information on the approval of the study protocol must also be provided in the manuscript.